Master in Computational Colour and Spectral Imaging (COSI)

Coloured Illumination Constancy in Intrinsic Image Decomposition via Cross-Render Albedo Invariance (CARI)

Master Thesis Report

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Abstract

Intrinsic image decomposition (IID) tries to separate a photograph into two layers: the *albedo*, which is the true colour of each surface, and the *shading*, which captures how light falls on those surfaces. Most IID models assume that light is white or achromatic. In real indoor scenes, however, light sources have colour: warm lamps cast orange tints, while windows let in cool blue daylight. This coloured illumination becomes entangled with the recovered albedo. As a result, the same painted wall looks like it has different colours depending on which light is on, even though the paint itself has not changed.

In this work we tackle this coloured illumination problem directly. We propose *Cross-Render Albedo Invariance* (CARI), a training strategy that uses pairs of images of the same scene captured under different illuminants. The core idea is simple: if the same surface appears in two photographs lit differently, its albedo must be identical in both. We enforce this with a loss that penalises any change in the predicted albedo across the two views, while requiring that the shading layer absorbs the difference.

We implement CARI inside a model built on a frozen DINOv2-Large encoder and a DPT-style decoder. A key practical insight is that standard white-balance preprocessing, commonly applied to training data, removes the very illuminant colour information that CARI needs to function; we therefore train on raw, unprocessed image pairs.

Evaluating this claim turned out to require correcting the way it is measured. The chroma-cast score used throughout this literature, and in earlier drafts of this work, pools albedo chromaticity across all the materials of a scene before taking its variance, and therefore conflates the illuminant-induced drift it intends to measure with the ordinary chromatic diversity of the scene. Because that second term is computed from the *prediction*, a model can lower the score simply by washing the colour out of its albedo. The metric pays models to destroy the very quantity the task exists to recover. We decompose it into a scale-normalised invariance term and a fidelity term anchored on the pseudo-ground-truth albedo, and show that the correction reverses the ranking of our own ablation.

We evaluate on four benchmarks spanning real and synthetic scenes: MID, IIW, MAW and ARAP. Under the corrected metric our model is the only one in the comparison that is simultaneously colour-faithful and competitively invariant: it reproduces the between-material chroma of the pseudo-ground-truth albedo closely (a fidelity of 0.941, against 0.728 for the strongest diffusion baseline and 0.484 for the strongest feed-forward one) and attains the best material-lightness stability in the roster, while remaining within a fifth of the most illuminant-invariant method on hue. We do not claim the invariance crown: two external methods hold it, and we show that both purchase it by discarding roughly a quarter and a half of real albedo colour respectively. The same degenerate trade that our own colour-skip ablation makes, which is visible to the naked eye. These results are obtained with 18.5M trained parameters and a single forward pass, some seventy times fewer trained weights and five times faster than the diffusion baselines. The remaining gap is in perceptual shading quality measured by IIW, which we analyse through the constancy–structure tension inherent to IID.

Acknowledgment

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I also thank 3DUniversum for providing the computing resources, funding, and supportive research environment that made this work possible.

This project builds on the work of the intrinsic image decomposition community. I am grateful to the researchers who made their publications, datasets, and implementations publicly available. The code and supporting material for this thesis are available at `https://github.com/tmkhang1999/CARI`.

Finally, this thesis marks the completion of my Master's studies in the COSI Erasmus Mundus programme. I thank the COSI board for its continued support and for enabling me to carry out this research in collaboration with 3DUniversum.

Glossary

Albedo	The intrinsic material colour of a surface, independent of the illumination. A white wall has high albedo in all three colour channels; a black wall has low albedo. In the IID model $\mathbf{I} = \mathbf{A} \cdot S$, the albedo is the term $\mathbf{A}$.
Chromaticity	The colour of a pixel or region expressed independently of its overall brightness. For an RGB pixel (r, g, b), the chromaticity is typically the normalised vector $(r, g, b)/(r+g+b)$. A constancy metric based on chromaticity measures whether the hue of the predicted albedo shifts when the illuminant changes.
Colour cast	A global or regional shift in the hue of an image caused by a coloured light source. A warm orange lamp produces an orange cast; a fluorescent tube produces a green cast. When a cast leaks into the predicted albedo it means the model has not separated the illuminant colour from the material colour.
Illuminant	The light source or combination of light sources illuminating a scene. An illuminant is characterised by its spectral power distribution, which determines both its brightness and its colour (e.g. warm, cool, neutral).
Intrinsic Image Decomposition (IID)	The task of decomposing a photograph into an albedo layer (material colour) and a shading layer (lighting effect). See Chapter 3 for the formal definition.
Diffuse shading	The per-pixel quantity that encodes how much light falls on each surface point, combining the effect of surface normal orientation, light distance and direction, and occlusion (shadows). In this work the shading is a three-channel (colored) quantity $\mathbf{S}_d$, so it can represent a coloured illuminant; see Chapter 3.

Non-diffuse residual	The remaining image content not explained by the albedo-times-diffuse-shading product: specular highlights, visible light sources, and clipped regions. Computed analytically as $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$, with no dedicated network head.
White-balance (WB)	An image processing step that neutralises the colour of the illuminant, making images appear as if they were lit by a pure white light. White balance is applied by cameras automatically or in post-processing. In this work, WB is intentionally *disabled* for the MID training pairs because it removes the illuminant colour information that the CARI training strategy relies on.
Scale-and-shift invariance (SSI)	A property of a metric or loss that makes it insensitive to global multiplicative and additive offsets. An SSI shading loss aligns the predicted shading to the ground truth using the best global scale and bias before computing the error, so it measures the quality of the relative shading pattern rather than its absolute level.

Acronyms

IID	Intrinsic Image Decomposition
CARI	Cross-Render Albedo Invariance
MID	Multi-Illumination Dataset
ARAP	As Real As Possible (evaluation dataset)
MAW	Measured Albedo in the Wild
IIW	Intrinsic Images in the Wild
ViT	Vision Transformer
DPT	Dense Prediction Transformer
SSI	Scale-and-Shift Invariant
LMSE	Local Mean Square Error
si-RMSE	Scale-Invariant Root Mean Square Error
WHDR	Weighted Human Disagreement Rate
RCI	Reflectance and Chromaticity Inconsistency
RMSE	Root Mean Square Error
L1	Absolute Error (mean absolute deviation)
MSG	Multi-Scale Gradient
DSSIM	Structural Dissimilarity
WB	White Balance

Notations

$\mathbf{I} \in [0,1]^{H\times W\times 3}$	Observed RGB image
$\mathbf{A} \in [0,1]^{H\times W\times 3}$	Albedo (intrinsic material colour)
$\mathbf{S}_d \in \mathbb{R}_{>0}^{H\times W\times 3}$	Colored diffuse shading (three-channel)
$\mathbf{R} \in \mathbb{R}_{\geq 0}^{H\times W\times 3}$	Non-diffuse residual; $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$
$\boldsymbol{\pi} \in (0,1)^{H\times W\times 3}$	Network shading output before inversion; $\mathbf{S}_d = (1 - \boldsymbol{\pi})/\boldsymbol{\pi}$
$(\mathbf{I}_1, \mathbf{I}_2)$	A cross-render pair: same scene photographed under two different illuminants
Ω_v	Set of valid paired pixels (excludes clipped/near-black pixels in either frame of a pair)
Ω_m, Ω_d	Set of pixels with valid ground-truth labels; set of pixels with diffuse-shading ground truth
$\nabla\mathbf{A}$	Spatial gradient of the albedo
$L(\cdot)$	Rec. 601 luminance of an RGB quantity
$\mathcal{L}_{\text{a}}$	Albedo ℓ_1 supervision loss
$\mathcal{L}_{\text{dssim}}$	Albedo structural-dissimilarity loss
$\mathcal{L}_{\text{msg-a}}$, $\mathcal{L}_{\text{msg-s}}$, $\mathcal{L}_{\text{msg-r}}$	Multi-scale gradient losses (albedo, shading, reconstruction)
$\mathcal{L}_{\text{chroma}}$	Albedo chroma-direction loss (hue/saturation, brightness-independent)
$\mathcal{L}_{\text{s}}$	Scale-and-shift invariant (SSI) shading loss
$\mathcal{L}_{\text{recon}}$	Diffuse reconstruction loss; $\lVert\mathbf{A} \odot \mathbf{S}_d - \mathbf{A}^* \odot \mathbf{S}_d^*\rVert_1$
$\mathcal{L}_{\text{res}}$	Residual closure and sparsity loss
$\mathcal{L}_{\text{mat}}$	DINOv2-token material-consistency loss
$\mathcal{L}_{\text{inv}}$	Cross-render invariance loss (CARI, albedo consistency)
$\mathcal{L}_{\text{expl}}$	Shading explanation loss (CARI, illuminant ratio)
$\mathcal{L}_{\text{sign}}$	Shading sign loss (positivity prior)
$\lambda_\star$	Weight of the corresponding loss term $\mathcal{L}_\star$
C_{mat}	MID chromaticity constancy metric (lower is better)
C_{arap}	ARAP constancy metric (lower is better)
Cast_{RMS}	RMS colour cast residual (lower is better)
WHDR	Weighted Human Disagreement Rate on IIW (lower is better)
ΔE	CIELAB colour difference on MAW (lower is better)

Contents

1 | Introduction

A photograph does not record what things look like. It records how light from those things reached the camera at one specific moment.

after Edwin Land

When we look at a red apple under a warm yellow lamp, we still see a red apple. Our visual system separates the colour of the apple's skin from the colour of the light hitting it. This ability, called *colour constancy*, is so automatic in humans that we rarely think about it. For computers, it is a hard problem. A camera simply records the product of surface colour and illumination, and the two are mixed together in every pixel. Separating them is the goal of *Intrinsic Image Decomposition* (IID).

IID has a long history in computer vision and colour science, dating back to the Retinex theory of Land and McCann (1971). The core idea is that a photograph $\mathbf{I}$ can be modelled as the product of two layers: an *albedo* $\mathbf{A}$ and a *shading* S.

$$\mathbf{I}(p) = \mathbf{A}(p) \cdot S(p), \qquad \forall\, p \in \Omega \tag{1.1}$$

Here p denotes a pixel, Ω is the set of all pixels, the albedo encodes the true material colour at each pixel, and the shading encodes how bright the surface is due to lighting and geometry. If we can recover $\mathbf{A}$ reliably, we obtain a lighting-independent description of the scene. This separation has practical consequences. An interior-design tool can recolour a wall without painting over its shadows; an augmented-reality system can place a virtual chair under the room's existing light; a photograph can be re-lit without changing the apparent material; and a catalogue or inspection system can compare product colours captured under different lamps. All four applications fail when lamp colour remains baked into the estimated material.

Observed image $\mathbf{I}$ | Material albedo $\mathbf{A}$ | Diffuse shading $\mathbf{S}_d$ | Non-diffuse residual $\mathbf{R}$

Figure 1.1: *Visual meaning of the decomposition used in this thesis. The photograph is explained by material colour multiplied by the illumination arriving at each surface, plus a residual for effects such as highlights and visible light sources. These are illustrative predictions, not separate photographs.*

1.1 The Coloured Illumination Problem

Most IID methods, both classical and modern, share an implicit assumption: the illuminant is grey. Under a grey illuminant, the shading S in Eq. (1.1) is a single scalar value per pixel, and colour information in the image comes entirely from the albedo. This simplification allows the model to be trained with standard supervision, and it works well when the light is close to white.

Real indoor scenes, however, rarely have pure white light. Incandescent bulbs emit warm orange-yellow light; fluorescent tubes lean green; LEDs can be cool blue; afternoon sunlight through a window can have a strong orange cast. When illumination is coloured, the shading is no longer achromatic because it carries a hue, and the simple product model in Eq. (1.1) breaks down. What typically happens is that the model absorbs the illuminant colour into the albedo, because the albedo is the only 3-channel (RGB) term in the equation. The result is that the same wall looks orange when lit by one lamp and blue when lit by another, even though its paint has not changed.

This is not a failure of any specific model; it is a fundamental ambiguity. Without additional constraints, there is no way to tell from a single image whether the orange hue in a pixel comes from orange paint or orange light. The state of the art handles this by training on large synthetic datasets with diverse illumination, which provides statistical regularisation, but it does not resolve the core ambiguity. Figure 1.2 shows a concrete example: two photographs of the same indoor scene lit by different coloured lamps. A model trained without illuminant-invariance constraints assigns different albedos to the same wall in each image, while our approach, detailed in this report, keeps the predicted albedo stable.

Figure 1.2: *Albedo predictions on one MID scene under four light conditions. The upper row shows the four inputs and the dataset pseudo-ground-truth albedo; the lower row shows the four predictions and a per-pixel coefficient-of-variation map (red marks regions that change with the light). The predicted albedo stays consistent as the illumination changes, so the CoV maps remain cool over the flat surfaces where the colour cast varies most: the constancy this thesis targets.*

1.2 Our Approach: Cross-Render Albedo Invariance

We attack the coloured illumination problem through a training-time constraint we call *Cross-Render Albedo Invariance* (CARI). The idea is straightforward: if a dataset contains images of the same physical scene photographed under different coloured illuminants, then a correct IID model must produce the same albedo for all of those images. We can enforce this with a loss term that directly penalises any change in the predicted albedo across the illuminant conditions. At the same time, a companion loss ensures that the shading layer absorbs and explains the difference between the two images, so the model cannot simply predict a flat albedo and ignore the image content.

To apply CARI in practice, we need a dataset with real multi-illuminant pairs. We use the Multi-Illumination Dataset (MID), which contains indoor scenes photographed under several different light sources. A crucial practical insight arises here: most datasets pre-process images with *white balance* (WB), which neutralises the illuminant colour and produces images that all look as if they were lit by white light. This pre-processing is counterproductive for our goal. CARI needs the illuminant colour to be visible in the images so the model can learn to separate it from the material colour. We therefore train on raw, un-white-balanced MID pairs, preserving the full illuminant information in the training signal.

We implement CARI inside a model built on a frozen *DINOv2-Large* vision encoder (Oquab et al., 2024) and a Dense Prediction Transformer (DPT) decoder (Ranftl et al., 2021). The model is first trained on Hypersim (Roberts et al., 2021), a large photorealistic synthetic dataset with ground truth albedo and shading, which gives it a strong baseline decomposition. CARI losses are then added in a second stage, using the MID raw pairs to push the model toward illuminant-invariant albedo predictions.

1.3 Contributions

The main contributions of this work are:

- **Cross-Render Albedo Invariance (CARI).** We introduce paired invariance and explanation losses that require the same scene to retain its albedo while shading explains the change of illumination.
- **A raw-pair training result.** Controlled ablations show that white balancing removes much of the colour variation needed by CARI; retaining raw MID pairs materially improves the targeted constancy measures.
- **A corrected evaluation of colour constancy.** We show that a pooled chroma-cast score can reward desaturated albedo, then separate cross-light invariance from calibration to a pseudo-albedo reference. Four complementary benchmarks report real constancy, measured colour, synthetic dense accuracy, and perceptual reflectance ordering.
- **3D-Front-IID and an informative negative result.** We render 5,056 paired observations with deliberately varied illuminant colour and exact albedo. The corpus improves the colour axis it covers but not ARAP's hard-shadow failure mode, exposing a specific gap in the renderer rather than hiding a null result.
- **Optimisation evidence.** The final refinement study documents the measured trade-off between invariant material colour, hard-shadow leakage, and retained image structure.

The project code, training configurations, and evaluation scripts are publicly available at `https://github.com/tmkhang1999/CARI`.

The rest of this report is organised as follows. Chapter 2 reviews related work on IID, multi-illuminant scenes, and the datasets used for evaluation. Chapter 3 describes the model architecture and the CARI training strategy. Chapter 4 covers the loss terms, training data, and optimisation schedule in detail. Chapter 5 presents quantitative and qualitative experiments on all four benchmarks, together with the ablation studies and a controlled negative result. Chapter 6 demonstrates qualitative factor edits built directly on the decomposition: residual suppression and factor-space illumination adjustment. Chapter 7 summarises the findings, discusses limitations, and outlines directions for future work.

2 Related Work

The history of science is the history of struggles against the default assumption.

Anonymous

This chapter reviews prior work that is relevant to the approach taken in this thesis. We cover classical and learning-based IID methods in Sections 2.1 and 2.2, work on multi-illuminant scenes and colour constancy in Sect. 2.3, and the evaluation benchmarks we use in Sect. 2.4.

2.1 Classical Intrinsic Image Decomposition

The problem of separating illumination from reflectance is rooted in Retinex theory (Land and McCann, 1971). Its starting observation is that perceived lightness depends strongly on spatial comparisons, not only on the number recorded at one pixel. If neighbouring image intensities are $I_1 = A_1 S_1$ and $I_2 = A_2 S_2$, their ratio is

$$\frac{I_1}{I_2} = \frac{A_1}{A_2}\frac{S_1}{S_2}. \tag{2.1}$$

When illumination varies slowly over that small neighbourhood, $S_1/S_2 \approx 1$, so the observed ratio is evidence for a reflectance change. For example, a painted black stripe on a white wall should remain a material boundary as the room becomes brighter. A soft illumination gradient should not. Retinex-style methods classify or threshold such local changes and propagate the surviving ratios to estimate a reflectance field.

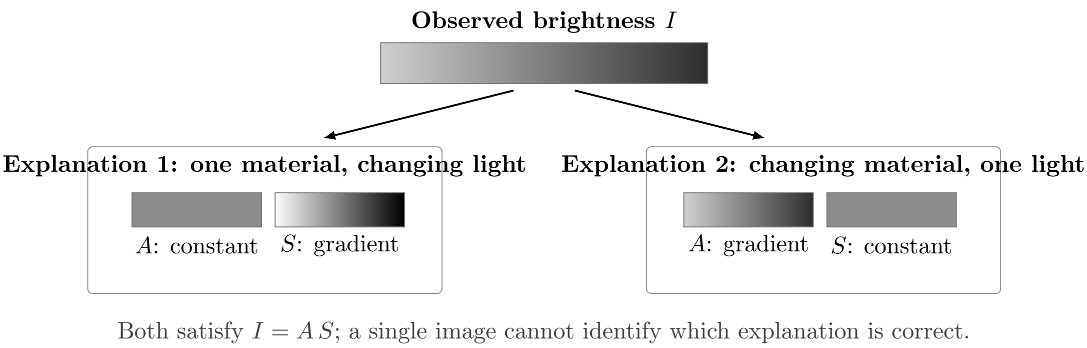

Figure 2.1: *The intrinsic-image ambiguity in one dimension. The same observed brightness gradient can be explained by constant reflectance under changing illumination (left), or by changing reflectance under constant illumination (right). Retinex and learned IID methods resolve this underdetermined factorisation by adding spatial, semantic, and data-derived priors.*

This is foundational because it turns an impossible per-pixel factorisation into a problem that can use spatial context. It is also only a prior. A hard cast shadow can change abruptly and resemble paint; textured material can vary rapidly; specular highlights violate the diffuse product model. The common shorthand "illumination is low-frequency and reflectance is high-frequency" is therefore a useful intuition, not a physical rule. Modern learned methods replace many hand-designed thresholds with data, but they still rely on the same distinction between material changes and lighting changes.

Figure 2.1 shows why such assumptions are necessary. Even a simple brightness gradient has multiple exact decompositions: the image alone does not reveal whether the change came from the surface or the light.

Grosse et al. (2009) introduced the MIT Intrinsic Images dataset, the first public benchmark for IID, and used it to show that simple Retinex baselines outperformed more sophisticated methods of the time on real photographs. A consistent finding from this benchmark is that the problem is extremely ill-posed: many different (albedo, shading) pairs explain the same image. Without strong priors, models tend to find trivial solutions.

The most influential approach to adding structure to the ill-posed problem is Shape, Illumination, and Reflectance from Shading (SIRFS) by Barron and Malik (2015). SIRFS frames IID as an energy minimisation over albedo, shading, and surface normals jointly. It encodes strong priors: albedos should be piecewise-constant, shadings should be smooth, and surface normals should form a plausible scene geometry. SIRFS significantly outperformed previous methods and remained a strong baseline for several years. However, its hand-designed priors struggle on complex real-world scenes with cluttered backgrounds, unusual lighting, or materials that violate the piecewise-constant albedo assumption.

2.2 Learning-Based Intrinsic Image Decomposition

The availability of large synthetic datasets shifted the field toward learned models. Li and Snavely (2018a) proposed CGIntrinsics, a set of physically-based re-renderings of synthetic indoor scenes that provides paired (image, albedo, shading) training data at scale. They showed that a CNN trained on this data, combined with ordinal supervision from the human judgements of IIW (Bell et al., 2014), generalises well to real photographs. CGIntrinsics established the pattern that has dominated the field since: train on photorealistic synthetic data with ground-truth labels, then evaluate on real images.

Bell et al. (2014) introduced the Intrinsic Images in the Wild (IIW) benchmark, which takes a different approach to ground truth. Rather than rendering or measuring true albedos, IIW collects human judgements about relative albedo comparisons: for any two surface patches in a photograph, human annotators say which one is lighter. This produces a set of relative constraints that any valid albedo prediction must satisfy. The evaluation metric, the Weighted Human Disagreement Rate (WHDR), measures how often the model's albedo ordering disagrees with the majority of human annotators. IIW became the standard benchmark for IID for many years and is still widely used. Its limitation is that it captures perceptual shading quality (whether shadows look right) more than it captures the accuracy of the recovered material colour.

A line of work by Careaga and Aksoy is the closest to ours in its treatment of illumination colour. Ordinal Shading (Careaga and Aksoy, 2023) decomposes real photographs by first estimating shading up to an ordinal (ordering) constraint, which is far more robust than absolute regression, and then refining the albedo; its shading is still greyscale. Its successor, Colorful Diffuse IID (CD-IID) (Careaga and Aksoy, 2024), is the first method to explicitly predict a *colored* diffuse shading in the wild, through a multi-stage cascade (grey decomposition, then a low-resolution chromaticity network, then albedo and diffuse-shading refinement) and an analytic non-diffuse residual. CD-IID is the direct point of comparison for the goal of this thesis: it shows that colored shading is representable, but its training never *constrains* the albedo to be invariant across illuminant changes. Consequently, the colour split between albedo and shading is learned only implicitly from single images. Our work adds exactly that missing constraint, using real cross-illuminant pairs.

In parallel, diffusion-based models have become strong IID baselines. Marigold-

IID (Ke et al., 2025) fine-tunes a Stable-Diffusion backbone for IID and ships two variants: an *appearance* model trained on InteriorVerse that targets material properties (albedo, roughness, metallic), and a *lighting* model trained on Hypersim (Roberts et al., 2021) that targets albedo, diffuse shading, and a non-diffuse residual. Both produce visually sharp decompositions and excellent zero-shot WHDR on IIW. We use Marigold-IID as our primary experimental baseline because checkpoints are public, its lighting variant shares our model's training distribution (Hypersim), and it represents the current generative state of the art. Very recent work reaches for external knowledge instead: ReasonX (Dirik et al., 2025) guides the decomposition with a multimodal language model and reports strong IIW WHDR improvements; no public checkpoint was available at the time of writing, so we cite its published numbers only where protocols match.

Comparison scope.. The methods above are not all compared in the same way. We run models locally only when an official checkpoint and compatible output are available: Marigold-IID, CRefNet, and Ordinal Shading. We cite CD-IID, ReasonX, and older methods from their papers only when the dataset split and metric match our protocol. This distinction matters because a published number obtained with different training data, resizing, or test scenes is context, not a controlled head-to-head result. Chapter 5 marks locally reproduced and literature-cited rows separately and explains why each model is included.

The fundamental limitation shared by these methods is that none of them explicitly *supervises* the separation of illuminant colour from material colour. When the scene is lit by a coloured light, the illuminant hue can bleed into the predicted albedo, because nothing in the training objective says it should not. Our work addresses this gap with a constraint mined from multi-illuminant data.

2.3 Multi-Illuminant Scenes and Colour Constancy

The specific challenge of coloured illumination in IID has received relatively little attention compared to the broader IID problem. Most methods implicitly assume a single white or near-white illuminant.

The cost of that assumption is arithmetic rather than model-specific. If shading is scalar, it has no hue, so a coloured illuminant must be represented in the only RGB factor: albedo. Chapter 3 derives this limitation and demonstrates it visually in Figure 3.1; this section focuses on how prior work has responded to it.

Work on multi-illuminant colour constancy in computational photography provides useful context. Beigpour et al. (2014) addressed the problem of estimating multiple coloured illuminants from a single image using a conditional random field formulation. They created the MIID dataset, which contains scenes photographed under combinations of coloured light sources with ground-truth illuminant annotations. This is a different task from IID (they estimate illuminant colours rather than separate albedo from shading), but the setup is closely related and provides inspiration for the multi-illuminant pair training we use in this work.

The closest prior work to the constraint we propose is the self-supervised decomposition network of Kinoshita and Kiya (2021). They observe the same gap we do: most IID methods do not enforce any consistency on the *reflectance* layer, and the usual white-illuminant model cannot express a coloured light in the first place. Their answer is to adopt a colour-illuminant model and to train with losses computed over several illumination conditions of the same content, so that the predicted reflectance is pushed to agree across them. The difference from our work lies in where those conditions come from. Their illumination variants are *simulated*, which makes the supervision cheap but ties it to the realism of the simulator. We instead mine the constraint from real photographs of real rooms under real bounced light (Sect. 2.4.2), where the illuminant chromaticity varies for physical reasons rather than by construction. The measurements in Chapter 5 show why this distinction matters: a synthetic tint applied to an image is not distributed like a real illuminant, and training on the synthetic version measurably hurt constancy.

This family of work also suggests the shape of a sensible metric. If a model is genuinely illuminant-invariant, the chromaticity it assigns to one physical material must not move when the light changes. Our constancy measures (C_{mat} on MID, C_{arap} and $\mathrm{Cast}_{\mathrm{RMS}}$ on ARAP) are built directly on that statement: they group the renderings or photographs of one scene by illuminant condition and report how much the predicted albedo drifts across the group (Sect. 5.2). They are internal diagnostics rather than a published protocol, which is why we report them alongside, not instead of, the measured-albedo benchmarks.

2.4 Evaluation Benchmarks

We evaluate our model on four benchmarks that together cover different aspects of IID quality.

Dataset, split, protocol, and benchmark.. These terms are kept distinct throughout the thesis. A *dataset* is a collection of images and annotations. A *split* identifies which samples are used for training, validation, or testing. An evaluation *protocol* specifies preprocessing, eligible samples, and a metric. A *benchmark* is the held-out data together with that protocol. Thus MID is both a source of training pairs and an evaluation dataset, but the CARI training scenes and the 30-scene test split never overlap. Calling a collection a "database" adds no useful distinction here, so this thesis uses *dataset* consistently.

The four benchmarks answer different questions. MID tests whether a real surface stays stable across lights; MAW measures real material colour on sampled regions; ARAP provides dense synthetic albedo and repeated illumination; IIW tests whether predicted relative reflectance agrees with people. No single score is treated as a complete definition of IID quality.

2.4.1 IIW

Intrinsic Images in the Wild (Bell et al., 2014) is the most widely used benchmark in the IID literature. It contains 5,230 real indoor and outdoor photographs collected from the internet, annotated with human pairwise reflectance comparisons. The evaluation metric is WHDR: the fraction of annotator pairs (weighted by confidence) with which the model's albedo ordering disagrees. Lower WHDR is better. IIW rewards agreement with perceived relative reflectance. Its human labels are appropriate for that perceptual question, but the ordinal lightness comparison does not measure RGB colour, absolute albedo, or dense texture. We evaluate on the standard test split of 1,046 images (every fifth image of the 5,230), and report WHDR as the mean over images rather than pooled over judgements.

2.4.2 MID: Multi-Illuminant Dataset

MID, the Multi-Illumination Dataset of Murmann et al. (2019), is the dataset most directly relevant to the goal of this work. It contains over 1,000 real indoor scenes, each photographed 25 times from a fixed viewpoint with a flash bounced in a different direction for every shot. Because the bounced light picks up the colour of the wall or ceiling it reflects from, the *effective illuminant chromaticity varies across the 25 conditions* (we measured a 15–25% chromaticity spread on the dataset's mirror-probe annotations), while the scene and camera stay pixel-aligned. The key property is that the true albedo of each surface does not change across the illuminant conditions; only the illuminant does. This makes it possible to define a constancy metric (C_{mat}) that measures how much the scale-normalised predicted albedo lightness changes across conditions. A model with perfect illuminant invariance would achieve $C_{\mathrm{mat}} = 0$. We hold out 30 scenes as a test split and report constancy on those; the training split is used for the CARI pairs (Chapter 4).

2.4.3 ARAP

ARAP ("As Real As Possible") is a synthetic high-dynamic-range dataset of around 50 rendered indoor and outdoor scenes (Bonneel et al., 2017). A subset of scenes ships with multiple renderings under different lighting conditions (direction, intensity, and in some scenes light colour); we evaluate constancy on the 37 multi-light scene groups (23 indoor), and additionally report the coloured-illuminant subset in which the light *colour* varies across renderings, which is the axis this thesis targets. ARAP provides ground-truth albedo images, which allows measuring not only constancy across conditions (the C_{arap} and Cast_{RMS} metrics) but also the accuracy of the predicted albedo against the ground truth. The constancy metrics group scene renderings by the illuminant condition and measure how consistent the predicted albedo is across groups. Our local study reports LMSE, SSIM, and a mean-normalised RMSE diagnostic. This implementation is not the least-squares scale-invariant RMSE used in published ARAP tables, so Chapter 5 keeps the local and published protocols separate. The two types of metric capture different things: a model can have good constancy (stable albedo across illuminants) while still being wrong about the absolute material colour, and vice versa.

2.4.4 MAW: Measured Albedo in the Wild

MAW (Wu et al., 2023) is a benchmark of real indoor images paired with ground-truth albedo measurements obtained through a controlled physical measurement process. It provides a direct measure of albedo colour accuracy through two metrics: ΔE (the CIELAB colour difference between predicted and measured albedo) and intensity SI-MSE (scale-invariant mean square error on the luminance channel). ΔE captures colour error in a perceptually motivated space after the official scorer's per-material scale alignment. MAW is particularly valuable because it measures real scenes where the illuminant colour is part of the challenge, making it complementary to the synthetic ARAP. We evaluate on 874 images from the MAW test split.

3 Method

This chapter describes how our model works. We start with the physical model that underlies IID (Sect. 3.1), then describe the network architecture (Sect. 3.2), and finally explain the CARI training strategy (Sect. 3.3).

3.1 Physical Model

3.1.1 The Standard IID Equation

Under the Lambertian reflectance model with a single distant light source, the intensity at a pixel p in an image is:

$$\mathbf{I}(p) = \mathbf{A}(p) \cdot S(p) \tag{3.1}$$

where $\mathbf{A}(p) \in [0, 1]^3$ is the three-channel albedo (material reflectance) and $S(p) \in \mathbb{R}_{>0}$ is a non-negative scalar shading value. Shading describes the illumination arriving at the surface point after geometry, visibility, and shadows have had their effect. Albedo describes the fraction and colour of that light that the material diffusely reflects. The camera observes the reflected result, not either factor in isolation.

Under the assumption of a grey (achromatic) illuminant, S is a scalar and the product in Eq. (3.1) is a simple element-wise multiplication: each colour channel of $\mathbf{I}$ is scaled by the same amount. This makes the colour of each pixel determined entirely by $\mathbf{A}$.

3.1.2 Why Achromatic Shading Fails with Coloured Light

When the illuminant has colour, the scaling is no longer the same across channels. A red lamp makes the R channel brighter relative to G and B. A correct model would encode this in the shading, but scalar shading cannot express different scalings

Figure 3.1: *Why scalar shading must leak illuminant colour into albedo. From left: a warm-lit MID frame; pseudo-ground-truth albedo; the RGB shading implied by the data; and the albedo recovered with the luminance of that true shading. The favourable scalar construction still produces a visibly tinted albedo (mean chroma error* 2.31*, versus* 0.22 *when shading has three channels), because a scalar can alter brightness but not hue. The remaining RGB error comes from clipping and the display-domain conversion, not from an inability to represent channel-dependent illumination.*

per channel. What actually happens in practice is that the model absorbs the channel imbalance into the albedo, because the albedo is the only RGB term in the equation. The predicted albedo therefore reflects the combined colour of the surface *and* the illuminant, rather than just the surface.

Figure 3.1 demonstrates that this is a limitation of the representation, not a particular network. Even when the scalar model is given the luminance of the true RGB shading, dividing the image by that scalar can change brightness but cannot remove the lamp's hue.

To model coloured illumination correctly, the shading must therefore be a three-channel (RGB) quantity $\mathbf{S}_d(p) \in \mathbb{R}^3_{>0}$, so each colour channel can be scaled independently. The model additionally keeps an explicit non-diffuse term for the light that the Lambertian product cannot represent (specular highlights, visible light sources, clipped pixels), giving the full decomposition:

$$\mathbf{I}(p) = \mathbf{A}(p) \odot \mathbf{S}_d(p) + \mathbf{R}(p) \tag{3.2}$$

where $\odot$ is element-wise multiplication, $\mathbf{S}_d$ is the colored *diffuse* shading, and $\mathbf{R} \geq 0$ is the non-diffuse residual. The model predicts $\mathbf{A}$ and $\mathbf{S}_d$ with two network heads and derives the residual *analytically* as $\mathbf{R} = \left(\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d\right)_+$, so the reconstruction $\mathbf{A} \odot \mathbf{S}_d + \mathbf{R}$ matches the input by construction wherever the diffuse product does not overshoot it.

A three-channel shading solves the representation problem but not the *ambiguity* problem: single-image supervision cannot say how much of a hue belongs to the albedo and how much to the shading colour, and the standard ground-truth datasets (including Hypersim) supervise shading only up to this ambiguity. This is where

CARI comes in: by constraining the albedo to be identical across images of the same scene under different illuminants, the colour split stops being a free choice: any illuminant colour that the model puts into the albedo now violates the invariance constraint, so it is forced into $\mathbf{S}_d$, which is exactly where the physics says it belongs.

3.1.3 The Shading Parameterisation

A practical challenge in predicting shading is that it can take any positive value, which is an unbounded output range; worse, its distribution in real scenes is heavily skewed. Specular surfaces and bright light sources produce very large shading values, so linear shading concentrates almost all pixels in a narrow band near zero with a long tail (Figure 3.2): a regression loss on this representation is dominated by a few extreme pixels while the mid-range values that carry the shadow structure receive almost no gradient. Working in the logarithmic domain balances the distribution but still has no well-defined range for a network output. Following the inverse-shading parameterisation of Careaga and Aksoy (2023), which was introduced to solve exactly this conditioning problem, the network predicts a bounded quantity $\boldsymbol{\pi}(p) = 1/\big(\mathbf{S}_d(p) + 1\big) \in (0, 1)$ per channel using a sigmoid activation, and the shading is recovered by inverting this mapping:

$$\mathbf{S}_d(p) = \frac{1 - \boldsymbol{\pi}(p)}{\boldsymbol{\pi}(p)} \tag{3.3}$$

The prediction is clamped to $\boldsymbol{\pi} \in [5 \times 10^{-3}, 1 - 10^{-4}]$, which bounds the recovered shading to $\mathbf{S}_d \in [10^{-4}, 199]$, providing enough headroom to represent bright high-dynamic-range illumination while keeping the division numerically safe. The inversion is differentiable, so gradients flow through it during training. Losses that supervise the shading pattern are computed on $\boldsymbol{\pi}$ (bounded, well-conditioned), while physics losses that need the actual light intensity (reconstruction, the residual) use the linear $\mathbf{S}_d$.

3.2 Network Architecture

Our model is built on two components: a frozen DINOv2-Large visual encoder and a DPT-style decoder with separate output heads for albedo and shading.

3.2.1 DINOv2 Encoder

The encoder is DINOv2-Large (Oquab et al., 2024), a Vision Transformer (ViT) with a patch size of 14 pixels and approximately 304 million parameters. DINOv2

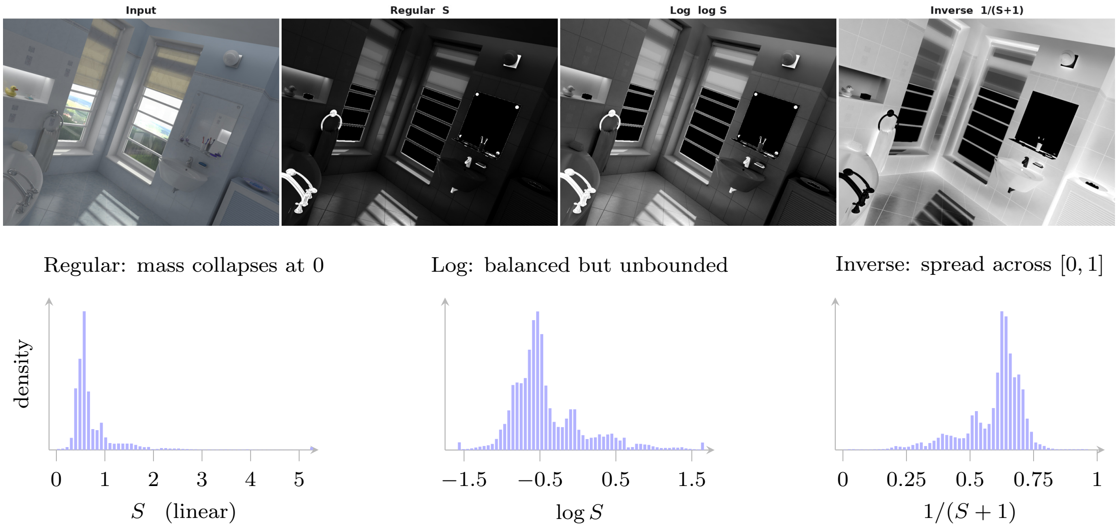

Figure 3.2: *Why shading is regressed in the inverse domain. Shading is derived from Hypersim ground truth as $S = \mathbf{I}/\mathbf{A}$ and shown in three representations, with the distribution of its values beneath each. Linear shading is dominated by a small number of very bright pixels. It runs to 5.3 on this frame, so almost all of the mass collapses into the first bin and a regressor spends its capacity on outliers. Log shading balances the distribution but has no bounded range, so it cannot be produced by a saturating output nonlinearity. Inverse shading, $1/(S+1)$, is confined to $[0, 1]$* by construction *and spreads across it, which makes it a well-conditioned target for a sigmoid head: this is the representation our shading branch predicts (Sect. 3.1), following Careaga and Aksoy (2023). The physics losses that need real light intensity operate on the linear $\mathbf{S}_d$ recovered from it.*

was trained with self-supervised learning on a large and diverse image corpus, and it produces patch-level feature representations that capture rich semantic and geometric information.

We use DINOv2 in a frozen state, meaning its weights are not updated during IID training. This is a deliberate design choice for two reasons. First, DINOv2's representations are already very strong; fine-tuning them would require careful learning rate scheduling and risks destroying useful features learned during self-supervised pre-training. Second, keeping the encoder frozen reduces GPU memory during training, which allows us to use a larger batch size.

For an input image of size $H \times W$, DINOv2 divides it into $(H/14) \times (W/14)$ patches and produces a token embedding for each patch. We extract the intermediate features from four roughly evenly-spaced transformer blocks (blocks 5, 12, 18, and 24 of the 24-block ViT-L, the last of which is the final block). These multi-depth features capture information at different levels of abstraction: earlier blocks retain more local texture and colour, while later blocks encode semantic and structural information. One consequence of DINOv2's heavily augmented self-supervised training is directly useful for IID: because its objective is invariant to strong photometric augmentation, its patch tokens are expected to be robust to low-level appearance changes including illumination, so a shadowed and a lit patch of the same material tend to map to similar tokens, which is a helpful prior for assigning the same albedo across a shadow boundary.

3.2.2 DPT Decoder

The Dense Prediction Transformer (DPT) decoder (Ranftl et al., 2021) was designed to reconstruct full-resolution feature maps from the patch tokens of a ViT. The key idea is to re-assemble the patch tokens from each layer into a spatial feature map, then fuse the maps from all layers through a series of convolutional feature fusion blocks. The fusion progressively upsamples the features to the input resolution, combining coarse semantic information from deep layers with fine spatial details from early layers.

We use a single DPT decoder that takes all four feature levels as input. Each level is first *reassembled*: the patch tokens are projected and reshaped into a spatial feature map at a level-specific resolution. The four maps are then merged by a chain of *fusion* blocks, each of which upsamples the coarser map, adds the next finer one, and refines the sum with residual convolutions, so semantic context from deep blocks and spatial precision from early blocks accumulate into one representation. Because the ViT operates on 14×14 patches, even the finest token map is far below pixel resolution; a lightweight convolutional *detail stem* therefore processes the raw input image in parallel and injects genuine high-frequency edge information that the token path cannot carry. The decoder output is a shared full-resolution feature map, which is fed to two separate output heads.

3.2.3 Output Heads

The albedo head takes the DPT feature map, concatenates a full-resolution *RGB skip* (the gamma-encoded input image), and passes the result through a small convolutional block followed by a sigmoid activation:

$$\mathbf{A} = \sigma\Big(\mathrm{Conv}\big(\mathbf{F}_{\mathrm{DPT}},\, \mathbf{I}^{1/2.2}\big)\Big) \in [0,1]^{H\times W\times 3} \tag{3.4}$$

The sigmoid bounds the albedo to $[0,1]^3$, the correct range for a reflectance value. The RGB skip is motivated by a practical risk of using a frozen semantic encoder:

its tokens need not retain all of the local, absolute colour information required for albedo prediction. We did not directly probe DINOv2 for illumination invariance, so this is a design hypothesis rather than an established property of the encoder. Feeding image colour directly into the head restores that capacity. Crucially, this skip is only safe *in combination with* the CARI invariance loss: with a direct colour path and no invariance constraint, the illuminant colour would flow straight through the skip into the albedo. This coupling is one of the ablation axes in Chapter 5.

The shading head applies its own convolutional block and sigmoid and outputs the three-channel inverse shading:

$$\boldsymbol{\pi} = \sigma\big(\mathrm{Conv}(\mathbf{F}_{\mathrm{DPT}})\big) \in (0,1)^{H\times W\times 3} \tag{3.5}$$

The colored diffuse shading $\mathbf{S}_d$ is recovered from $\boldsymbol{\pi}$ using Eq. (3.3), and the non-diffuse residual follows analytically:

$$\mathbf{R} = \big(\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d\big)_+ \tag{3.6}$$

There is no third head: the residual costs no parameters, and the reconstruction $\mathbf{A} \odot \mathbf{S}_d + \mathbf{R}$ is exact wherever the diffuse product does not overshoot the image.

Both heads share the same DPT feature map as input. This encourages the two predictions to be consistent with each other, since they are both derived from the same underlying scene representation.

3.3 Cross-Render Albedo Invariance (CARI)

3.3.1 The Core Idea

CARI is a training strategy, not a change to the network. It adds two loss terms that operate on *pairs* of images of the same scene photographed under different illuminants.

Let $(\mathbf{I}_1, \mathbf{I}_2)$ be such a pair, where $\mathbf{I}_1$ and $\mathbf{I}_2$ show the same scene under two different coloured light sources. Let $\mathbf{A}_1 = \mathbf{A}(\mathbf{I}_1)$ and $\mathbf{A}_2 = \mathbf{A}(\mathbf{I}_2)$ be the albedos predicted by the model for each image, and let S_1 and S_2 be the corresponding shadings.

3.3.2 Invariance Loss

The albedo must be the same for both images, because the physical material has not changed. We encode this as a masked ℓ_1 loss:

$$\mathcal{L}_{\mathrm{inv}} = \frac{1}{|\Omega_v|} \sum_{p\in\Omega_v} \big\|\mathbf{A}_1(p) - \mathbf{A}_2(p)\big\|_1 \tag{3.7}$$

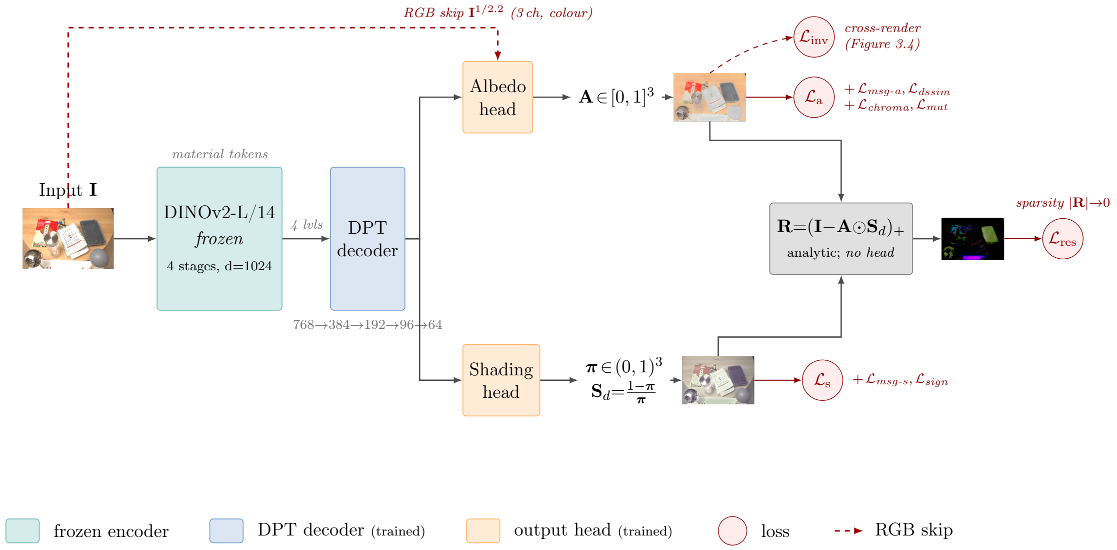

Figure 3.3: *Model architecture and its supervision. A* frozen *DINOv2-Large encoder (teal) extracts patch features from four transformer stages; because it is never fine-tuned it supplies the illumination-invariant material tokens the backbone was pre-trained to produce, and only* 18.5*M parameters downstream are learned. A DPT decoder (blue) reassembles and fuses the four levels into a shared full-resolution feature map. Two output heads (amber) read it. The albedo head additionally takes a full-resolution* RGB skip*: the gamma-encoded input image, concatenated in directly so the material colour DINOv2's illumination-invariant tokens discard is not lost. This skip is safe only because $\mathcal{L}_{inv}$ constrains what it can carry through; without that constraint the illuminant colour would flow straight down it into* $\mathbf{A}$*. The shading head has no such skip; it predicts the inverse shading $\boldsymbol{\pi}$ from the shared DPT features alone, from which $\mathbf{S}_d = (1 - \boldsymbol{\pi})/\boldsymbol{\pi}$ follows. The non-diffuse residual $\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$ is analytic; there is no third head. Red nodes are the training losses on each output: the albedo family ($\mathcal{L}_a$ with its multi-scale-gradient, structural, chroma and material-consistency terms), the shading family ($\mathcal{L}_s$ with its gradient and sign terms), and a sparsity term $\mathcal{L}_{res}$ that pushes the analytic residual toward zero. The cross-render invariance loss $\mathcal{L}_{inv}$ also constrains the albedo but acts across an illuminant pair at training time only (Figure 3.4). Thumbnails are this model's own outputs on a MID test scene.*

where $\Omega_v \subset \Omega$ is the set of *valid paired pixels*. The validity mask matters in practice: MID frames are high-dynamic-range captures that are clipped per frame, so a pixel can be saturated in one frame and not the other, and deep shadows carry mostly sensor noise. Ω_v excludes clipped and near-black pixels in either frame (roughly 75% of pixels survive the mask on average), so the invariance constraint is only applied where both observations are trustworthy. This loss is zero only when the model predicts identical albedo for both images. Minimising it forces the model to make the albedo independent of the illuminant colour.

3.3.3 Explanation Loss

The invariance loss alone is not enough. A degenerate model could satisfy Eq. (3.7) by predicting a flat, constant albedo (the same grey image for every input). To prevent this, we add a companion loss that requires the shading to *explain* the pixel-level difference between the two images.

Under the diffuse model $\mathbf{I} \approx \mathbf{A} \odot \mathbf{S}_d$, the ratio of corresponding pixels in the two images equals the ratio of their shadings wherever the albedo is shared:

$$\frac{\mathbf{I}_1(p)}{\mathbf{I}_2(p)} = \frac{\mathbf{S}_{d,1}(p)}{\mathbf{S}_{d,2}(p)}, \qquad \text{if } \mathbf{A}_1(p) = \mathbf{A}_2(p) \tag{3.8}$$

The material term cancels in the ideal diffuse image ratio, making the ratio a useful lighting cue wherever registration is valid and non-diffuse effects are small. Saturation, shadows, and specularities violate that idealisation; the validity mask limits rather than eliminates those failures. We enforce the constraint on the *luminance* of both ratios, in the log domain for numerical robustness:

$$\mathcal{L}_{\text{expl}} = \frac{1}{|\Omega_v|} \sum_{p \in \Omega_v} \left| \log \frac{L(\mathbf{I}_1(p)) + \varepsilon}{L(\mathbf{I}_2(p)) + \varepsilon} - \log \frac{L(\mathbf{S}_{d,1}(p)) + \varepsilon}{L(\mathbf{S}_{d,2}(p)) + \varepsilon} \right| \tag{3.9}$$

where $L(\cdot)$ is the Rec. 601 luminance, $\varepsilon = 10^{-3}$, and Ω_v is the same validity mask as in Eq. (3.7). Using luminance makes this term supervise the achromatic *intensity* change between the frames. Colour separation is produced by the interaction of the three-channel decomposition, albedo/reconstruction supervision, and $\mathcal{L}_{\text{inv}}$; the invariance term alone does not identify where the rejected colour must go. This loss is zero only when the shading ratio matches the image ratio at every valid pixel, which means the shading is absorbing the lighting difference rather than the albedo.

3.3.4 Why Raw (Un-White-Balanced) Pairs Are Necessary

A subtle but important point about applying CARI is the role of white balance. Standard image pre-processing in most datasets applies automatic white balance, which estimates the illuminant and rescales the colour channels so that neutral surfaces appear neutral. A common heuristic is the *grey-world* assumption: the average scene reflectance is treated as grey. It fails when the scene itself is not colour-balanced. For example, in a room dominated by a red painted wall, the average red signal can come from the material rather than the lamp; forcing that average to grey removes genuine wall colour. White balance is therefore an estimate, not ground truth. After white balance, the two images $\mathbf{I}_1$ and $\mathbf{I}_2$ of the same scene under different illuminants will look very similar in terms of global colour, because the illuminant colour has been removed.

This is precisely the wrong input for CARI. The luminance explanation loss in Eq. (3.9) needs the intensity change, whereas the colour variation is needed by the RGB invariance and reconstruction constraints. If both images are white-balanced to look neutral, much of that RGB variation is removed before training, so CARI can no longer teach the model which colour belongs to the illuminant rather than the surface.

We confirm this by measuring the chromaticity spread of the MID dataset with and without white balance. The ground-truth illuminant probes show 15–25% variation in chromaticity across the different illuminant conditions; this variation is almost entirely removed by white balance. We therefore set the `mid_raw_color_pair` flag to `true` in our training configuration, which disables white balance for MID pairs.

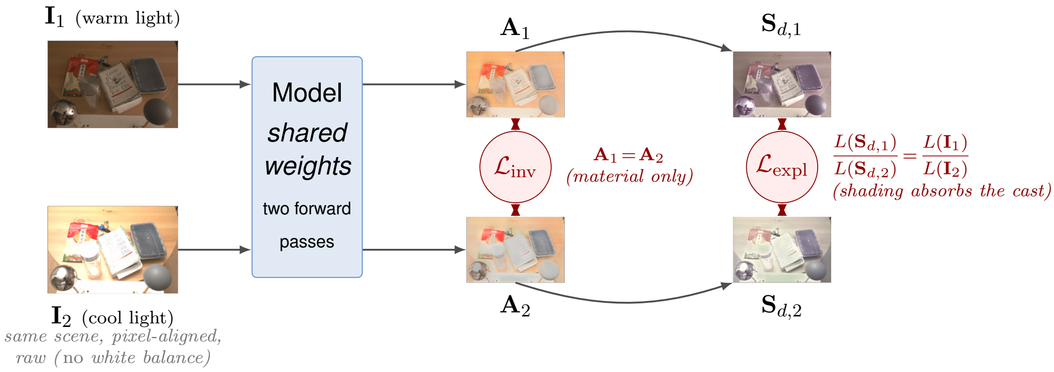

Figure 3.4: *Cross-Render Albedo Invariance (CARI), applied only at training time. A pixel-aligned pair of the same scene under different coloured illuminants is passed through the* same *model (shared weights) in two forward passes. The input thumbnails carry visibly different casts, yet the two predicted albedos $\mathbf{A}_1, \mathbf{A}_2$ are nearly identical, as enforced by $\mathcal{L}_{inv}$, while the predicted shadings $\mathbf{S}_{d,1}, \mathbf{S}_{d,2}$ differ, having absorbed the illuminant. $\mathcal{L}_{expl}$ makes that explicit: it forces the ratio of the two shadings to match the ratio of the two inputs, so any colour the albedo declines to absorb must be explained by the shading. White balance is disabled for these pairs (Sect. 3.3.4) so that the illuminant colour survives in the input ratio that $\mathcal{L}_{expl}$ reads. Thumbnails are this model's own outputs.*

4 Implementation

This chapter covers the full set of loss terms used during training (Sect. 4.1), the training datasets and how we prepare them (Sect. 4.2), and the optimisation schedule that controls which losses and datasets are active at each stage (Sect. 4.3). Every value in this chapter is taken directly from the training configuration used to produce the final model.

4.1 Loss Terms

Our model is trained with three families of losses: *supervised anchors* that tie the predictions to ground-truth labels on synthetic data, *CARI losses* that enforce illuminant invariance on real multi-illuminant pairs, and *physics and regularisation terms* that keep the decomposition consistent. The total loss is a weighted sum; Table 4.1 lists every active term with its weight.

4.1.1 Supervised Anchor Losses

Albedo ($\mathcal{L}_{\mathrm{a}}$, $\mathcal{L}_{\mathrm{dssim}}$, $\mathcal{L}_{\mathrm{msg\text{-}a}}$, $\mathcal{L}_{\mathrm{chroma}}$). The core albedo supervision is a masked ℓ_1 loss against the ground-truth albedo $\mathbf{A}^*$:

$$\mathcal{L}_{\mathrm{a}} = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \left\| \mathbf{A}(p) - \mathbf{A}^*(p) \right\|_1 \tag{4.1}$$

where Ω_m excludes invalid pixels (missing labels, clipped regions). This is the dominant *absolute-colour anchor* of the whole system: CARI constrains albedo to be equal *across* illuminants but says nothing about what the correct colour *is*; without a strong absolute anchor, the invariance constraint alone could be satisfied by grey. Three companions sharpen this anchor. A structural dissimilarity term $\mathcal{L}_{\mathrm{dssim}}$ (DSSIM, weight 0.2) penalises local structure differences. A multi-scale gradient term $\mathcal{L}_{\mathrm{msg\text{-}a}}$ (weight 0.5) matches predicted albedo gradients to ground-truth gradients across four dyadic scales, placing each edge where the ground truth

Table 4.1: *Loss terms used in training, with the weights of the final configuration. Supervised anchors require ground-truth labels and are gated to the synthetic datasets that provide them; CARI terms require multi-illuminant pairs and fire only on MID batches during the CARI stage; physics terms apply only where their required targets and masks exist. A gradient-decorrelation loss (forcing albedo and shading edges apart) was evaluated and* rejected*: it assumes albedo and shading edges never coincide, which is false in real scenes, and it erased genuinely shared edges.*

Loss term	Symbol	Weight	Applies to
Albedo ℓ_1	$\mathcal{L}_{\mathrm{a}}$	1.0	Hypersim, IV, MID (pseudo-GT)
Albedo DSSIM	$\mathcal{L}_{\mathrm{dssim}}$	0.2	same as $\mathcal{L}_{\mathrm{a}}$
Albedo multi-scale gradient	$\mathcal{L}_{\mathrm{msg\text{-}a}}$	0.5	same as $\mathcal{L}_{\mathrm{a}}$
Albedo chroma direction	$\mathcal{L}_{\mathrm{chroma}}$	0.2	same as $\mathcal{L}_{\mathrm{a}}$
Shading SSI (π-domain)	$\mathcal{L}_{\mathrm{s}}$	1.0	Hypersim (GT shading)
Shading multi-scale gradient	$\mathcal{L}_{\mathrm{msg\text{-}s}}$	0.5	Hypersim
Diffuse reconstruction	$\mathcal{L}_{\mathrm{recon}}$	1.0	Hypersim
Reconstruction gradient	$\mathcal{L}_{\mathrm{msg\text{-}r}}$	0.25	Hypersim
Residual closure + sparsity	$\mathcal{L}_{\mathrm{res}}$	0.5 / 0.02	Hypersim, MID
Material consistency (DINO)	$\mathcal{L}_{\mathrm{mat}}$	0.05	all datasets
Cross-render invariance	$\mathcal{L}_{\mathrm{inv}}$	0.5	MID raw pairs
Shading explanation	$\mathcal{L}_{\mathrm{expl}}$	0.25	MID raw pairs
Shading sign prior	$\mathcal{L}_{\mathrm{sign}}$	0.05	all datasets

says it belongs. Finally, a *chroma-direction* term (weight 0.2),

$$\mathcal{L}_{\mathrm{chroma}} = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \left\| \frac{\mathbf{A}(p)}{\|\mathbf{A}(p)\|} - \frac{\mathbf{A}^*(p)}{\|\mathbf{A}^*(p)\|} \right\|_1, \tag{4.2}$$

penalises hue and saturation error independently of brightness. This term exists because desaturation is the cheap escape for a pixel-wise loss: predicting a slightly grey version of every colour lowers ℓ_1 risk, and $\mathcal{L}_{\mathrm{chroma}}$ makes that escape expensive.

Shading SSI ($\mathcal{L}_{\mathrm{s}}$, $\mathcal{L}_{\mathrm{msg\text{-}s}}$). Shading is only defined up to an arbitrary global scale: a dimmer light with a brighter material explains the same image. The renderer's absolute light intensity varies freely between scenes, so a naive regression to ground-truth shading would spend most of its gradient fighting a quantity that is not identifiable. The standard remedy, introduced for the analogous scale ambiguity in monocular depth estimation (Ranftl et al., 2022) and adopted for shading by Careaga and Aksoy (2023), is a *scale-and-shift-invariant* (SSI) loss: before the error is computed, the prediction is aligned to the ground truth with the per-image scalar scale $\hat{s}$ and shift $\hat{t}$ that minimise the squared error, a two-parameter least-squares problem with a closed-form solution, so the alignment adds no optimisation difficulty. The aligned prediction is then penalised with a masked mean-squared error in the bounded $\boldsymbol{\pi}$ domain:

$$\mathcal{L}_{\mathrm{s}} = \frac{1}{|\Omega_d|} \sum_{p \in \Omega_d} \big(\hat{s}\, \boldsymbol{\pi}(p) + \hat{t} - \boldsymbol{\pi}^*(p)\big)^2, \qquad (\hat{s}, \hat{t}) = \arg\min_{s,t} \sum_{p \in \Omega_d} \big(s\, \boldsymbol{\pi}(p) + t - \boldsymbol{\pi}^*(p)\big)^2 \tag{4.3}$$

Ω_d is the *diffuse mask*: shading ground truth exists only on Hypersim, so this loss is gated to Hypersim batches. This supervises the relative shading pattern (shadows, gradients) without being confused by per-scene brightness. A multi-scale gradient companion (weight 0.5) sharpens shading boundaries. The SSI alignment is enabled after a 3,000-iteration warm-up, during which the un-aligned loss stabilises the output range first.

Diffuse reconstruction ($\mathcal{L}_{\mathrm{recon}}$, $\mathcal{L}_{\mathrm{msg\text{-}r}}$). Because the residual is analytic ($\mathbf{R} = (\mathbf{I} - \mathbf{A} \odot \mathbf{S}_d)_+$), a naive reconstruction loss $\|\mathbf{A} \odot \mathbf{S}_d + \mathbf{R} - \mathbf{I}\|$ would be identically zero wherever the diffuse product undershoots the image, which is a tautology that provides no gradient. We therefore supervise the *diffuse product* directly:

$$\mathcal{L}_{\mathrm{recon}} = \frac{1}{|\Omega_d|} \sum_{p \in \Omega_d} \big\| \mathbf{A}(p) \odot \mathbf{S}_d(p) - \mathbf{A}^*(p) \odot \mathbf{S}_d^*(p) \big\|_1 \tag{4.4}$$

The term is enabled only for Hypersim, where the target is the measured ground-truth diffuse product. It is a *two-sided* coupling: both predicted factors must jointly explain $\mathbf{A}^* \odot \mathbf{S}_d^*$. InteriorVerse and MID do not activate this loss in the implemented configuration; those batches are supervised by their available albedo, material, residual, sign, and CARI terms instead. This masking is important: defining an input-derived target for datasets without measured shading would change the trained objective.

Residual closure ($\mathcal{L}_{\mathrm{res}}$). The analytic residual creates an escape hatch: the model could dump hard-to- explain image content (for example a saturated red surface) into $\mathbf{R}$ instead of the albedo. We close it with a supervised residual block (weight 0.5): on Hypersim the residual target is the true non-diffuse component of the rendering, and on MID the target is $\mathbf{R}^* = 0$ away from specular pixels (physically true for the bounced-flash captures), so the loss there penalises *any* residual energy. A small sparsity term (weight 0.02) additionally pushes $\mathbf{R} \to 0$ inside the valid mask, so the diffuse product rather than the residual must explain the image.

4.1.2 CARI Losses

The two CARI losses $\mathcal{L}_{\text{inv}}$ (weight 0.5) and $\mathcal{L}_{\text{expl}}$ (weight 0.25) are defined in Sect. 3.3. They are computed by running the network a second time on the paired frame $\mathbf{I}_2$ within the same training step (a Siamese second forward with shared weights) and are masked by the intersection of the loss mask, the material-invariance mask, and the per-pair HDR validity mask. They fire on batches that contain MID raw pairs, i.e. from the CARI stage onwards (Sect. 4.3).

4.1.3 Physics and Regularisation Losses

Shading sign prior ($\mathcal{L}_{\text{sign}}$). The analytic residual is clamped non-negative, so any pixel where the diffuse product *overshoots* the image is un-modelled negative energy, which is something the decomposition cannot represent. The sign prior penalises exactly that overshoot:

$$\mathcal{L}_{\text{sign}} = \frac{1}{|\Omega_m|} \sum_{p \in \Omega_m} \left\| \max\big(\mathbf{A}(p) \odot \mathbf{S}_d(p) - \mathbf{I}(p),\, 0\big) \right\|_1 \tag{4.5}$$

The intuition is "anything that brightens the image beyond the light must come from the material, not the shading": the term pushes brightening texture out of the shading and into the albedo, complementing CARI (which pushes illuminant colour out of the albedo and into the shading).

Material consistency ($\mathcal{L}_{\text{mat}}$). The frozen DINOv2 tokens provide a label-free material signal: two patches with very similar tokens are very likely the same material. The material-consistency term (weight 0.05) penalises albedo chroma differences between patch pairs whose DINOv2 similarity exceeds a threshold, gently flattening the albedo *within* a material while leaving material boundaries untouched. Since the tokens are computed anyway for the encoder, this supervision is essentially free.

4.2 Training Data

Four datasets participate in training, each with a distinct role. In this chapter, *dataset* means a collection from which training samples are drawn. It does not mean an evaluation procedure. Chapter 5 uses held-out data together with a fixed protocol and metric; that combination is a *benchmark*. MID appears in both roles, but its training and test scenes are disjoint. Three are existing public corpora and carry the two-stage schedule of Sect. 4.3: Hypersim and InteriorVerse supply supervised albedo, and MID supplies the real cross-render pairs that CARI trains on. The fourth, 3D-Front-IID (Sect. 4.2.4), is rendered for this thesis to supply the one combination none of the others offers: a large, deliberately coloured illuminant change *and* a ground-truth albedo for the same camera. It enters at the refinement stage, where Sect. 5.7 studies its contribution in an exploratory configuration comparison.

4.2.1 Hypersim

Hypersim (Roberts et al., 2021) is a large photorealistic synthetic dataset of 461 indoor scenes generated with a physically-based renderer, providing ground-truth albedo (diffuse reflectance) and diffuse illumination for every image. It is the primary source of supervised signal: albedo, shading, reconstruction, and residual targets all come from Hypersim labels. The full release is large (multiple terabytes of EXR renders); under a local storage budget of 500 GB we download and train on approximately one quarter of the scenes, sampled across the scene split so that the material and layout diversity is preserved. This subset is sufficient because Hypersim's role is to anchor absolute albedo colour and supply shading ground truth, not to cover every scene; the cross-illuminant signal that the thesis depends on comes from the real MID pairs (Sect. 2.4.2), not from Hypersim. We use the standard scene split within the subset and hold out a validation set for monitoring and checkpoint selection.

4.2.2 InteriorVerse

InteriorVerse (Zhu et al., 2022a) is a rendered indoor dataset with ground-truth material, geometry, and spatially varying lighting annotations. It enters training as *albedo-supervision diversity*: more scenes, layouts, and materials for the absolute-colour anchor. Its shading is defined only implicitly ($\mathbf{S}_d^* = \mathbf{I}/\mathbf{A}^*$), so it is excluded from both shading-GT and diffuse-reconstruction losses in the implemented configuration. InteriorVerse contributes albedo supervision and material diversity, rather than a separately measured diffuse-image target.

4.2.3 MID: Real Multi-Illuminant Pairs

MID (Murmann et al., 2019) provides the cross-render training signal: over 1,000 real indoor scenes, each captured 25 times with a flash bounced in a different direction, pixel-aligned, with the 30 test scenes held out (Sect. 2.4.2). Each MID training sample contributes two things:

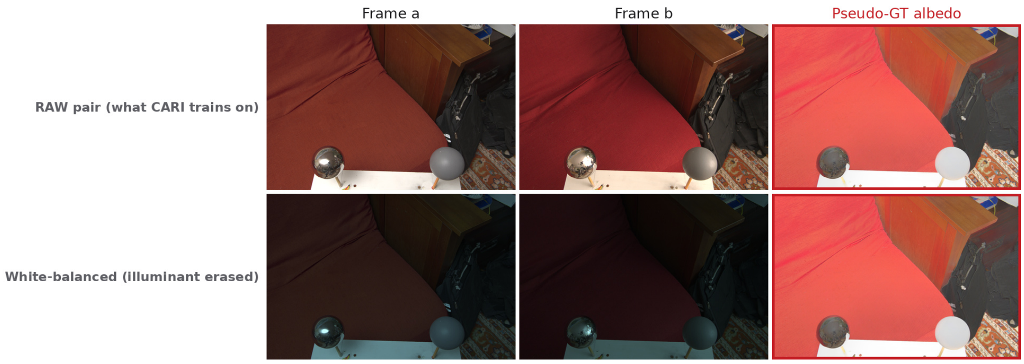

Figure 4.1: *MID preprocessing used during training; this is a data-loading diagram, not an evaluation protocol. The supervised path uses a white-balanced primary frame and the dataset pseudo-ground-truth albedo to anchor absolute material colour on real photographs. The CARI path keeps two aligned frames raw, without white balance, and masks pixels that are clipped or too dark in either exposure before applying the cross-render losses. This split keeps the albedo target stable while preserving the illuminant chromaticity change that CARI needs as supervision.*

(i) A supervised primary frame. The MID Intrinsics extension of Careaga and Aksoy (2023) supplies a dense pseudo-ground-truth albedo per scene, derived from the white-balanced multi-illumination stack; it is not part of the original MID release. The white-balanced primary frame is supervised against this pseudo-GT with the standard albedo losses, which anchors absolute colour on real photographs.

(ii) A raw cross-render pair. A second frame of the same scene, under a different bounce direction, is loaded *raw*, without white balance, together with a per-pair HDR validity mask that drops pixels clipped or near-black in either frame. The CARI losses operate on this pair. As established in Sect. 3.3, disabling white balance here is essential (the `mid_raw_color_pair` flag): the measured 15–25% illuminant chromaticity variation across bounce directions *is* the training signal, and white balance would erase it. An earlier configuration approximated this signal with synthetic chromatic augmentation (random global tints on the pair frame); it measurably hurt constancy on ARAP, because the synthetic tint distribution does not match real illuminants, and it was replaced by the raw pairs. Figure 4.1 summarises this preprocessing split: the white-balanced primary frame provides the supervised colour anchor, while the raw paired frames preserve the illuminant change used by CARI.

4.2.4 3D-Front-IID: A Rendered Cross-Illuminant Corpus

Motivation: the gap MID cannot close. MID is the only *real* source of cross-render pairs available to CARI, and the thesis depends on it, but it constrains the method along two axes at once. First, MID varies the illuminant by bouncing a single flash in a different *direction*: what changes most between two frames is the geometry and intensity of the light, and only incidentally its colour, whose chromaticity moves by roughly 15–25% across bounce directions (Sect. 2.4.2). The literal claim of this thesis is about *coloured* illumination, and the one dataset that supervises it does so only weakly and indirectly. Second, MID has no measured dense albedo: its pseudo-ground-truth is itself derived from the white-balanced capture stack. On a MID pair the invariance loss therefore acts without an independent absolute anchor on the same pixels, so nothing in the pair itself prevents the two predictions from agreeing on a *wrong* albedo; the anchor has to arrive from a different dataset, on a different image. The synthetic corpora have the opposite problem: Hypersim and InteriorVerse supply exact albedo, but each image is a single illumination, so they carry no cross-render signal at all.

Among the public datasets available to this project, none offered both properties together. We therefore rendered one.

How it is built. 3D-Front-IID is rendered from 3D-FRONT (Fu et al., 2021), a corpus of professionally designed indoor layouts furnished with 3D-FUTURE assets, using Blender's Cycles path tracer at 512×512 and 64 samples per pixel (`scripts/render_3dfront_dataset.py`). Four design decisions define it.

- **Interior camera sampling.** The layouts ship with exterior "dollhouse" camera poses, which are unusable as photographs. Cameras are instead sampled *inside* each room and accepted by ray casting a grid through the frustum: a view is kept only if it is not embedded in geometry, is not staring at a blank wall or the ceiling, and actually frames furniture rather than empty floor.

- **Randomised coloured illuminants: the thesis axis.** Each accepted view is rendered under $K \in \{2, 3\}$ lighting variants from the identical camera. Every variant re-randomises a key light (a blackbody between 2500 K and 9000 K, or a saturated hue) together with a near-neutral fill, and a *minimum rg-chromaticity gap* is enforced between the key lights of a view. The illuminant colour is thus a controlled, guaranteed-present variable rather than a by-product: every pair the loader can draw is a genuine colour change, by construction.

- **Exact albedo, pixel-aligned.** After the lit passes, every material is converted to an emission shader and all lights are removed, so the renderer returns each surface's base colour directly. The result is a true ground-truth albedo registered pixel-for-pixel with every lit variant of that view.
- **Automatic quality control.** Each rendered view is validated for exposure clipping, black frames, valid-pixel coverage and framing, and failures are quarantined rather than trained on; 103 of 2,599 rendered views were dropped this way.

The resulting corpus contains 960 rooms and 2,496 accepted views, yielding **5,056 cross-illuminant pairs**. The illuminant separation is wide and guaranteed: the median rg-chromaticity distance between the two illuminants of a pair is 0.23, and no pair falls below the enforced floor of 0.055. Figure 4.2 shows samples together with the measured illuminant gamut and pair-separation distribution.

What the corpus is designed to test. A 3D-Front-IID sample is the only one in the mix where the cross-render pair and an exact albedo target coexist *on the same pixels*: the loader returns two lit observations, the ground-truth albedo, and a per-pair validity mask. Two things follow. The invariance loss and the supervised albedo anchor act on the same image, so the degenerate solution available on an unanchored pair, in which two predictions drift together onto a consistent but incorrect albedo, is directly penalised rather than left to a different dataset to correct. And because the varied quantity is illuminant *colour* by construction, CARI is supervised on the axis the thesis actually claims, instead of inferring it from a direction change that only partly implies it. Its shading, like InteriorVerse's, is defined only implicitly as $\mathbf{S}_d^* = \mathbf{I}/\mathbf{A}^*$, so 3D-Front-IID is excluded from the shading-ground-truth-gated losses; it contributes the albedo anchor and the CARI pair, not a shading target. Sect. 5.7 measures what this contributes by adding the corpus to the training mix as one arm of an exploratory comparison. The configurations differ in more than one lever, so this study motivates a conclusion about the combined training recipe rather than a single-cause estimate of the corpus.

4.3 Optimisation Schedule

Training proceeds in two stages, as illustrated in Figure 4.3.

Stage A: synthetic warm-up. The model first trains on Hypersim alone, after which InteriorVerse joins the mix. Only the supervised, physics, and regularisation

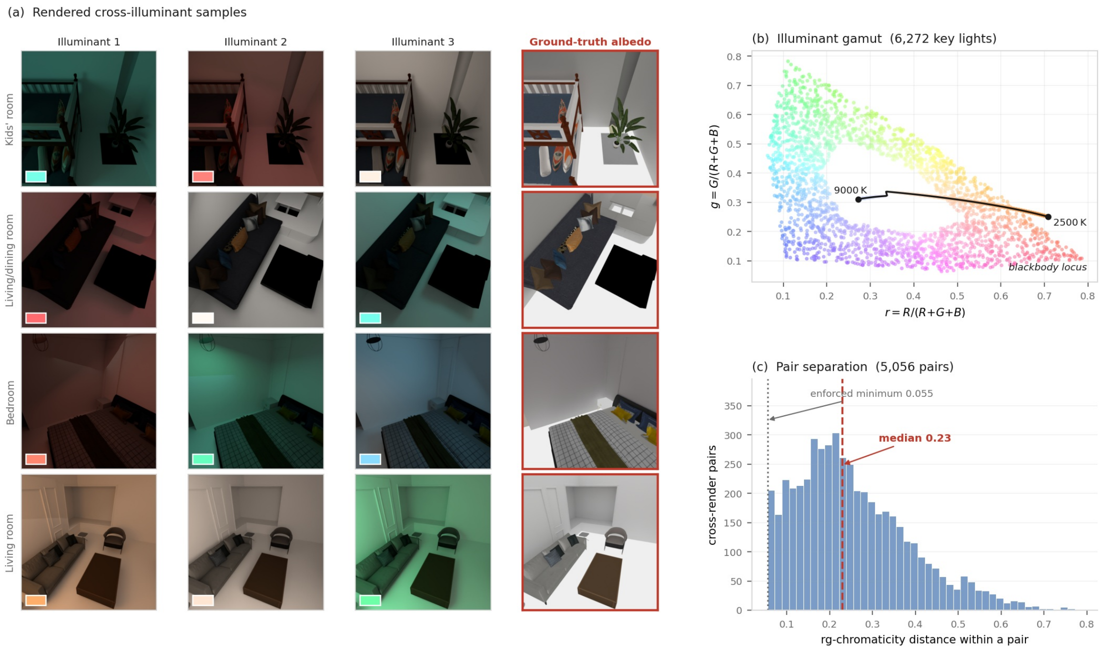

Figure 4.2: *3D-Front-IID, the cross-illuminant corpus rendered for this thesis. (a) Four rooms, each rendered from a fixed camera under the view's lighting variants (the swatch in each panel is that variant's key-light colour), shown beside the exact ground-truth albedo obtained from the emission pass. Each albedo is pixel-aligned with every lit variant. (b) The rg-chromaticity of all 6,272 key lights in the corpus, each drawn in its own colour, with the blackbody locus overlaid: sampling is deliberately not confined to the locus, so the illuminant gamut covers saturated colours that natural white-balanced capture never contains. (c) The chromaticity distance between the two illuminants of each of the 5,056 cross-render pairs. The renderer enforces a minimum separation, so every pair CARI can draw carries a real illuminant-colour change; the median separation is* 0.23.

losses are active. This stage gives the model absolute albedo colour and scale *before* any invariance constraint is applied. This ordering matters because, when applied to an unanchored model, an invariance loss can be satisfied by collapsing towards grey, whereas applied to a colour-anchored model it can only be satisfied by moving the illuminant colour into the shading. The Stage-A checkpoint is banked as the no-CARI baseline used throughout Chapter 5.

Stage B: CARI. Training resumes from the Stage-A checkpoint with the CARI configuration: the data mix becomes Hypersim / MID / InteriorVerse (with MID and InteriorVerse each roughly matching Hypersim's share), the raw cross-render pairs enter through MID, and $\mathcal{L}_{\mathrm{inv}}/\mathcal{L}_{\mathrm{expl}}$ activate. The synthetic majority keeps

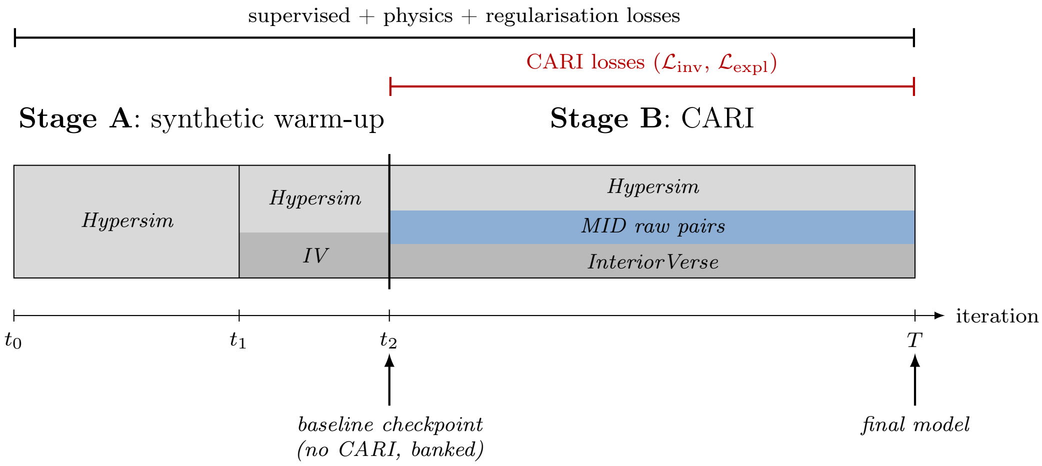

Figure 4.3: *Two-stage training schedule. Stage A establishes the supervised baseline on synthetic data (Hypersim, later joined by InteriorVerse). Stage B resumes from the Stage-A checkpoint and adds MID raw pairs with the CARI losses, keeping the synthetic majority as the absolute-colour anchor. The Stage-A checkpoint doubles as the no-CARI baseline in the ablation study (Sect. 5.6).*

re-anchoring absolute colour throughout, preventing drift while the invariance constraint reshapes the colour split. The end-of-schedule checkpoint is the final model reported in Chapter 5. We describe the schedule here by stage rather than by a specific iteration count, so that every configuration compared in Chapter 5 is held to the same curriculum and the same total training budget.

Optimisation parameters. We use Adam with a base learning rate of 2×10^{-4} and cosine annealing to 10^{-7} over the training horizon; the frozen DINOv2 encoder is excluded from the optimiser entirely. Batches of 4 crops at 512×512 are accumulated over 8 steps (effective batch size 32), with gradient-norm clipping at 1.0. All training runs on a single 24 GB NVIDIA Quadro RTX 6000.

The full-CARI configuration. The model reported as "full CARI" is this architecture trained under the two-stage schedule above to 50k iterations: Stage A establishes absolute albedo colour and shading structure from Hypersim and InteriorVerse, and Stage B adds the real MID raw cross-render pairs and activates the cross-render losses $\mathcal{L}_{inv}$ and $\mathcal{L}_{expl}$ together with the wide colour skip. Because the cross-render terms require a *second* forward pass on the paired frame, the CARI stage holds two graphs in memory; on the 24 GB card this caps the per-step crop count, which the gradient accumulation above compensates for. The frozen DINOv2-Large encoder (304 M parameters) is kept in evaluation mode throughout, so only the $\sim$18.5 M decoder-and-head parameters receive gradients, which is the main reason the whole pipeline fits on a single commodity GPU.

5 Experiments

This chapter reports results on the four benchmarks of Sect. 2.4, after the experimental setup in Sect. 5.1.

The order of the benchmark sections is deliberate, because the benchmarks do not carry equal weight and presenting them as a flat list would suggest that they do. We begin with MID (Sect. 5.2), the only benchmark that measures the property this thesis is about: whether the predicted albedo of a real surface stays put when the light on it changes. We then turn to IIW (Sect. 5.3), the field's long-standing perceptual benchmark. Its human judgements validly test relative reflectance ordering, but WHDR is blind to absolute albedo and hue. It is therefore complementary evidence rather than the deciding measure for coloured-illuminant leakage. That limitation is the reason MAW exists, so MAW follows (Sect. 5.4): it compares predictions against physically measured albedo on real surfaces and scores colour error directly, which is the closest thing the field has to an objective verdict on material colour. ARAP comes last (Sect. 5.5) because it is synthetic and therefore easiest to satisfy, but it is also the only benchmark with dense ground-truth albedo *and* repeated renderings of one scene under different lights, so it is the one place where constancy and accuracy can be measured on the same pixels.

All four benchmarks score the *albedo*, and that is deliberate rather than an oversight. The question this thesis asks is whether a surface's recovered colour survives a change of illuminant, so the albedo is the quantity under test; the shading is the layer into which the illuminant is displaced, and it is exercised qualitatively by factor-space edits in Chapter 6. We therefore do not report SAW (Careaga and Aksoy, 2023), which scores shading annotations: a good SAW score would say nothing about whether material colour is stable, which is the claim being made. This does mean that a decomposition could in principle satisfy every benchmark here while producing an implausible shading layer, and we do not rule that out. The application chapter demonstrates controllability, not a ground-truth shading evaluation.

Sect. 5.6 then isolates what CARI itself contributes, and Sect. 5.7 studies the final refinement choices.

5.1 Experimental Setup

Model configurations used in the results. Two headline configurations recur in the tables, and we use the following descriptive labels consistently throughout. **Ours (base CARI)** is the complete CARI configuration of Chapter 3, with both cross-render losses and the wide colour skip active at 40k iterations; it is Row 4 of the ablation in Sect. 5.6. **Ours (full model)**, our headline result, continues training for a further 20k iterations with the refinement of Sect. 5.7 (Row 4 of Table 5.8); it is the model meant wherever the text says "our model" unqualified. Generic phrases such as "our approach" refer to the method rather than a particular trained instance. The base-CARI row is carried alongside the full model so that the refinement's contribution is visible inside every table rather than folded silently into a single number. A third configuration, **Ours (IIW-fine-tuned)**, appears only where needed to diagnose the effect of direct IIW supervision and is never treated as a zero-shot result.

Baselines and how scores are obtained. The benchmark tables compare against the same roster of methods, and follow the convention used by (Careaga and Aksoy, 2024): a horizontal rule separates scores we *run locally* from scores we *cite* from the literature. We run Marigold-IID Appearance and Lighting (Ke et al., 2025), CRefNet (Luo et al., 2024), and the Ordinal Shading stage of CD-IID (Careaga and Aksoy, 2023) locally under our exact pipeline (zero-shot, same output-space normalisation and splits, identical inference resolution per benchmark). The full multi-stage CD-IID cascade (Careaga and Aksoy, 2024) and the older baselines are cited from their publications, and appear below the rule in each table. Our own constancy metrics (C_{mat}, C_{arap}, Cast_{RMS}) are not reported by those methods and are marked n/a; a cell marked "-" is simply one for which no value is available. Marigold-IID Lighting is trained on Hypersim, like our model, which makes it the most directly comparable external method.

Which published table each benchmark cites. The published comparison point differs per benchmark, chosen so that the cited protocol matches ours exactly rather than the nearest available number. MAW cites CD-IID Table 1: both the MAW authors' own evaluation code and our local harness apply the identical scaling and metrics. ARAP cites Ordinal Shading Table 1, *not* CD-IID Table 2: CD-IID modifies the ARAP scene set (dropping redundant illuminations and adding scenes from the MIST dataset), a set we cannot reproduce, whereas Ordinal Shading's table uses the original Bonneel et al. scenes (the same 52-scene set used throughout this chapter). IIW cites Ordinal Shading Table 2 for its zero-shot WHDR figure, alongside the fuller reproduced roster of (Wang et al., 2025). MID has no external citation on any axis; it is a ground-truth-free constancy metric that no other published work reports, and MIDIntrinsics is training data for both this model and CD-IID, which additionally disqualifies it as a zero-shot benchmark for either.

Methods compared. Table 5.1 summarises the methods along the axes that condition a fair reading of the benchmark numbers: architecture type, training data, parameter count, runtime, and native output space. Two points matter for interpretation. First, training data determines what counts as zero-shot. Our model trains on Hypersim, MID, and InteriorVerse, so it shares one source with each Marigold variant (Hypersim with Marigold-Lighting and InteriorVerse with Marigold-Appearance), which makes both directly comparable rather than only one. It is zero-shot on ARAP, MAW, and IIW, but in-domain on MID (it trains on the MID training split, disjoint from the test scenes), which is why the MID constancy metric is deliberately ground-truth-free (Sect. 5.2). CRefNet trains on the IIW training split, so its IIW WHDR is a non-zero-shot result (Sect. 5.3). Second, our model trains only about 18.5M parameters. The DINOv2-L encoder (304M) is frozen and excluded from the optimiser, so the trainable count is far lower trainable parameters than both the diffusion baselines and the multi-stage CD-IID cascade, an efficiency the constancy results should be read against.

Output-space normalisation. Marigold-IID's two variants use different output conventions: the Appearance variant outputs sRGB albedo, while the Lighting variant outputs linear albedo, shading, and residual layers. Before computing any metric, each prediction is converted to the corresponding linear albedo space; for sRGB predictions this applies the transfer function $A_{\text{lin}} = A_{\text{sRGB}}^{2.2}$. This correction matters because gamma encoding can artificially desaturate an albedo estimate and make it appear more constancy-correct than it actually is. All results in this chapter use this corrected normalisation.

Inference resolution. Inference resolution is fixed per benchmark rather than shared uniformly. MID, IIW, and ARAP use an aspect-preserving maximum side length of 1280 pixels; measured directly on the full ARAP set, 1280 and 1500 pixels give effectively identical scores (every metric moves by less than 1%, in mixed directions, consistent with measurement noise rather than a real effect), so 1280 is kept throughout for consistency and lower memory use. Ordinal Shading (Careaga and Aksoy, 2023) instead resizes each ARAP image to a content-adaptive resolution R_0 (Miangoleh et al.'s edge-density criterion), capped at 1500 pixels; since R_0 is computed per image and not reproducible from the published protocol alone, no fixed resolution can replicate it exactly, and the 1280-vs-1500 measurement above shows that choice is immaterial to our own scores regardless. MAW also uses the

Table 5.1: *Methods compared by architecture, training data, model size, inference cost, and native output. Runtime is the median of five timed runs after warm-up at 512×512 on one NVIDIA Quadro RTX 6000, using the same local interface; Marigold uses its recommended four denoising steps. Trainable parameters exclude frozen weights. A dash denotes a quantity not reported by the paper and not measured under our protocol. ReasonX's MLLM is a training-time judge, not the deployed intrinsic predictor.*

Method	Type	Training data	Train (M)	Total (M)	s/image	Output
Ours (full model)	DINOv2-L (frozen) + DPT, regression	Hypersim + MID + InteriorVerse	18.5	322.9	0.148	linear albedo
Marigold-App (Ke et al., 2025)	Stable-Diffusion latent diffusion	InteriorVerse	~1290	1290	0.805	sRGB albedo + material
Marigold-Light (Ke et al., 2025)	Stable-Diffusion latent diffusion	Hypersim	~1290	1290	0.982	linear albedo + shading + residual
CD-IID (Careaga and Aksoy, 2024)	Five-stage MiDaS-style cascade	8 synthetic sets + MIDIntrinsics; Hypersim for shading	~1685	~1685	–	linear albedo + RGB shading + residual
Ordinal Shading (Careaga and Aksoy, 2023)	CD-IID stage 1; ordinal MiDaS shading	CGIntrinsics + Hypersim + OpenRooms + GTA + MID	~337	~337	–	grayscale shading + implied albedo
PIE-Net (Das et al., 2022)	Edge-guided hybrid CNN	synthetic outdoor; MIT/IIW fine-tunes	204.1	204.1	–	reflectance + shading
CRefNet (Luo et al., 2024)	Decoder-sharing SwinV2 transformer	CGIntrinsics + IIW (+ NIID-Net pseudo-shading)	66.6	66.6	0.447	reflectance + shading
ReasonX (Dirik et al., 2025)	GRPO fine-tuning with InternVL2.5-4B judge	10k COCO RGB; judge: Hypersim + InteriorVerse	–	–	–	base-dependent intrinsic maps

Figure 5.1: *Effect of inference resolution on a MAW scene using the full-CARI model. The input photograph is shown on the left and the measured albedo on the right; the middle columns show predictions at different maximum side lengths. Each panel repeats the same black-sofa-armrest crop in the upper right with a red border to make blur differences easier to compare. Fine structure improves substantially from 512 to 1280 pixels, then largely plateaus. We therefore use the common 1280-pixel inference cap for every locally run MAW method; the official scorer subsequently reduces predictions to* 320×240 *before measuring the metrics.*

common 1280-pixel inference cap. Its public scorer then resizes every prediction and region mask to 320×240 before computing the metrics, so the reported numbers are determined at that fixed scoring scale. Every locally run method, including the baselines, is evaluated at the same resolution within each benchmark. Figure 5.1 shows the qualitative plateau this is chosen from: raising the full-CARI prediction from 1280 to 1792 or 2560 changes little visually, whereas 512 visibly softens fine structure at the resolutions where that matters (MID, IIW).

Benchmarks at a glance. Figure 5.2 previews the four evaluation benchmarks and the kind of ground truth each provides, which determines the metric it supports (Sect. 2.4). Figure 5.3 shows the qualitative comparison format used throughout this chapter: for a single in-the-wild image, the input and each method's predicted albedo side by side.

5.2 MID: Multi-Illuminant Constancy

MIDIntrinsics (Murmann et al., 2019) photographs the same real room 25 times, moving only the light, so it isolates the exact variable this thesis is about. Because it ships no independently measured dense albedo, it cannot tell us whether a prediction is *correct*; it can only tell us whether a prediction is *stable*. That makes it a constancy diagnostic rather than an accuracy benchmark, and we report it as such. We use the held-out 30-scene test split. The two scores follow the reflectance-consistency setting of (Kinoshita and Kiya, 2021) but are computed directly on MID's illuminant groups.

Both scores rest on the same fact: within one scene, the material at a given pixel is the same in every one of the 25 frames, so any variation in the predicted

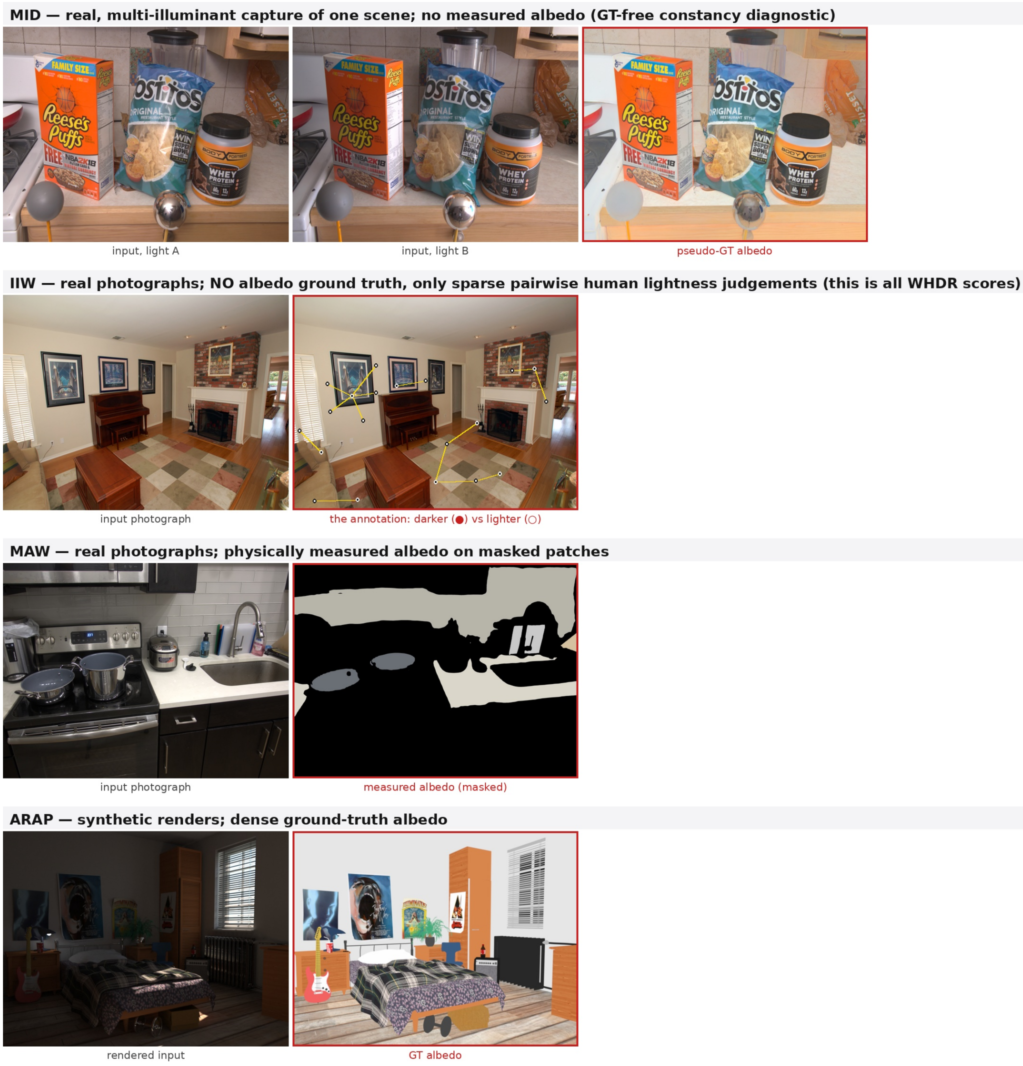

Figure 5.2: *The four evaluation benchmarks and the ground truth each provides, in the order the chapter presents them.* ***MID****: real multi-illuminant captures of one scene, with no measured albedo; which is why it supports a ground-truth-free constancy diagnostic and nothing stronger.* ***IIW****: real photographs with* no *albedo ground truth at all. The second panel overlays the annotation that does exist: sparse pairwise human lightness judgements, each a pair of points one of which was called darker (●) than the other (○). These sparse ordinal comparisons define what WHDR scores; they are informative about relative lightness but not RGB colour.* ***MAW****: real photographs with physically measured albedo on masked patches.* ***ARAP****: synthetic renders with dense ground-truth albedo.*

albedo across frames is error by definition. Let $k = 1, \ldots, N$ index the illuminant

Figure 5.3: *Qualitative albedo comparison on in-the-wild photographs (IIW), all methods run under the identical evaluation protocol. Each panel is scale-normalised by its own 99th percentile, because albedo is defined only up to a global scale and each method fixes that scale differently; beyond this, no colour correction is applied. The comparison is deliberately shown, but it cannot be adjudicated: IIW supplies no albedo ground truth (Figure 5.2), so an albedo that looks "cleaner" cannot be distinguished by eye from one that has simply had its colour removed. The diffusion baselines whiten these interiors considerably more aggressively than we do, and on this evidence alone one might read that as a better decomposition. The MID pseudo-GT calibration in Table 5.2 shows the ambiguity: Marigold-App and CRefNet retain* 72.8% *and* 48.4% *of the reference chroma spread, against our* 94.1%. *This figure shows method behaviour; it cannot establish correctness on IIW.*

conditions of a scene and let m index the material regions given by the scene's segmentation.

The first score asks whether the material's *lightness* holds still. Writing $\mu_{m,k}$ for the mean luminance of the predicted albedo over material m in frame k, we take the coefficient of variation across frames and average it over materials,

$$C_{\text{mat}} = \frac{1}{|M|} \sum_{m \in M} \frac{\operatorname{std}_k\left(\mu_{m,k}\right)}{\operatorname{mean}_k\left(\mu_{m,k}\right)}. \tag{5.1}$$

Dividing by the mean makes the score scale-free, which matters because albedo is only defined up to a global scale.

The second score asks whether the material's *colour* holds still, which is the axis a coloured lamp actually attacks. We reduce each material in each frame to its two chromaticity ratios $r_{m,k} = \bar{A}^R_{m,k}/\bar{A}^G_{m,k}$ and $b_{m,k} = \bar{A}^B_{m,k}/\bar{A}^G_{m,k}$. Normalising by the green channel removes brightness and leaves hue, so a model that lets a warm lamp tint its albedo will show drift in $r_{m,k}$ and $b_{m,k}$ even if its lightness is stable.

Why the pooled chroma spread is not a constancy metric. The natural first move is to pool these ratios over every material and every frame and report their combined spread,

$$\mathrm{Cast}_{\mathrm{RMS}} = \sqrt{\mathrm{Var}\big(r_{m,k}\big) + \mathrm{Var}\big(b_{m,k}\big)}, \tag{5.2}$$

where the variance runs over all (m, k) pairs jointly. This quantity is unsound as a measure of illuminant invariance, and because it is easy to arrive at we set out plainly why. By the law of total variance the pooled variance splits into

$$\underbrace{\mathrm{Var}_{m,k}\big(r_{m,k}\big)}_{\text{pooled}} = \underbrace{\mathbb{E}_m\big[\mathrm{Var}_k(r_{m,k})\big]}_{\text{within: drift across illuminants}} + \underbrace{\mathrm{Var}_m\big(\mathbb{E}_k[r_{m,k}]\big)}_{\text{between: chroma spread across materials}} . \tag{5.3}$$

Only the first term is the across-illuminant drift the metric is supposed to capture. The second is the chromatic diversity of the *scene*: a red sofa and a blue wall genuinely differ in hue, and this diversity enters the pooled score even under perfect illuminant invariance. It is tempting to dismiss it as a constant offset shared by every method evaluated on a fixed scene set, but it is not model-independent: the between-material term is computed from the *predicted* albedo, so a model that pulls distinct materials toward a common hue shrinks it, and thereby lowers Eq. (5.2) without becoming any more illuminant-invariant. **Pooling therefore rewards a model for destroying albedo colour.** This is not a hypothetical defect: in Sect. 5.6 the configuration that scores best under Eq. (5.2) is the one whose predicted albedo is visibly desaturated, and the configuration that best preserves the MID pseudo-GT colour calibration is scored worst.

Reported chroma metrics. We therefore report the two terms of Eq. (5.3) separately. Write the per-material drift and the between-material chroma spread of a prediction A as

$$\delta_m = \sqrt{\mathrm{Var}_k(r_{m,k}) + \mathrm{Var}_k(b_{m,k})}, \qquad S(A) = \sqrt{\mathrm{Var}_m\big(\bar{r}_m\big) + \mathrm{Var}_m\big(\bar{b}_m\big)}, \tag{5.4}$$

with $\bar{r}_m = \mathbb{E}_k[r_{m,k}]$ and $\bar{b}_m = \mathbb{E}_k[b_{m,k}]$. The constancy score is the mean drift measured against the chroma scale the model actually uses,

$$\mathrm{Cast}_{\mathrm{rel}} = \frac{\frac{1}{|M|}\sum_{m\in M}\delta_m}{S(A)}, \tag{5.5}$$

a coefficient of variation in chroma. The normalisation is necessary because δ_m is an absolute quantity: a model that carries twice as much albedo chroma has twice as much room to drift in absolute units, so comparing raw δ_m across models with different colourfulness penalises the more colourful one for being colourful. $\mathrm{Cast}_{\mathrm{rel}}$ instead asks the question that matters perceptually: whether two materials remain distinguishable when the lighting changes.

Invariance alone, however, can never be reported on its own, because a constant predictor attains it: a model that emits flat grey albedo is *perfectly* illuminant-invariant and scores zero on every drift measure. We therefore pair Eq. (5.5) with a calibration guard measured against MID's dataset-derived pseudo-ground-truth albedo. The primary guard is the mean per-material chromaticity error,

$$\mathrm{Chroma}_{\mathrm{err}} = \frac{1}{|M|} \sum_{m \in M} \left\| (\bar{r}_m, \bar{b}_m) - (\bar{r}^*_m, \bar{b}^*_m) \right\|, \tag{5.6}$$

the distance between each predicted material's chromaticity and its pseudo-GT chromaticity $(\bar{r}^*_m, \bar{b}^*_m)$; lower is better, and it is zero only when every material matches that reference, not merely its amount of colour. Alongside it we report a spread ratio,

$$\mathrm{Chroma}_{\mathrm{fid}} = \frac{S(A)}{S(A^*)}, \tag{5.7}$$

where A^* is the ground-truth albedo. This ratio is not a calibration score because a hue permutation preserving the spread would score 1. Instead, it is the specific detector of *collapse*: a model that greys its albedo drives $S(A)$, and hence $\mathrm{Chroma}_{\mathrm{fid}}$, well below 1, while an over-saturating model drives it above. The two answer different questions and we report both: $\mathrm{Chroma}_{\mathrm{err}}$ asks whether the colour is *right*, $\mathrm{Chroma}_{\mathrm{fid}}$ whether it has been *destroyed*. A low $\mathrm{Cast}_{\mathrm{rel}}$ is a result only when it comes with a low $\mathrm{Chroma}_{\mathrm{err}}$ and $\mathrm{Chroma}_{\mathrm{fid}} \approx 1$; achieved alongside colour collapse it is the degenerate solution, not constancy. (We report $\mathrm{Chroma}_{\mathrm{fid}}$ as the ratio of aggregate spreads rather than the mean of per-scene ratios; the latter is inflated by a small number of low-chroma scenes and is not a faithful summary.)

C_{mat} and $\mathrm{Cast}_{\mathrm{rel}}$ are lower-is-better and both vanish for a perfectly illuminant-invariant prediction. They are reported separately, rather than combined, because they can move in opposite directions: C_{mat} is a lightness score and $\mathrm{Cast}_{\mathrm{rel}}$ a hue score, and a configuration may stabilise one while disturbing the other.

This is deliberately an in-domain diagnostic: both our method and CD-IID train on the disjoint MID training split (Careaga and Aksoy, 2024). The table therefore compares locally evaluated models under one pipeline and makes no zero-shot claim.

Table 5.2 states the central result of this chapter, and it is a conjunction rather than a victory on any single column. Two facts hold simultaneously, and the paired bootstrap tells us which are real.

First, we do *not* win illuminant invariance, and we say so first. Marigold-App and CRefNet both attain $\mathrm{Cast}_{\mathrm{rel}} = 0.355$ against our 0.421, and the paired difference

Table 5.2: *Multi-illuminant constancy on the held-out MIDIntrinsics test split (Murmann et al., 2019), 30 scenes; brackets are 95% percentile-bootstrap intervals over scenes. C_{mat} (Eq. (5.1)) is lightness stability and $Cast_{rel}$ (Eq. (5.5)) is scale-normalised hue stability; lower is better for both. $Chroma_{err}$ (Eq. (5.6), lower is better) is the primary colour-calibration guard (the distance of each predicted material's chromaticity from pseudo-GT), and $Chroma_{fid}$ (Eq. (5.7), ratio of aggregate spreads, target 1) is the collapse detector: below 1 is greyed colour, above 1 over-saturation. Gold, silver, and bronze mark first, second, and third numerical rank in each column; $Chroma_{fid}$ is ranked by absolute distance from its target of 1, and displayed ties share a rank. The bootstrap intervals, rather than the colour fill, determine whether a difference is statistically supported. These are internal metrics, so no published method reports directly comparable values; the full CD-IID cascade is excluded because only its Ordinal Shading stage is integrated in our evaluation harness. † denotes the full model after IIW fine-tuning; this diagnostic row tests whether improving WHDR transfers to MID constancy and is not the selected model.*

Model	C_{mat} ↓	$\text{Chroma}_{\text{err}}$ ↓	Cast_{rel} ↓	$\text{Chroma}_{\text{fid}}$ (→ 1)
Ours (full model)	**0.130 [.117,.144]**	**0.121 [.102,.143]**	0.421 [.357,.485]	0.941
Ours (IIW-fine-tuned)†	0.191 [.172,.211]	0.134 [.115,.157]	0.426 [.360,.493]	0.978
Ours (base CARI)	0.157 [.141,.174]	0.129 [.109,.150]	0.425 [.360,.491]	**0.999**
Marigold-App (Ke et al., 2025)	0.193 [.173,.214]	0.195 [.152,.245]	**0.355 [.309,.406]**	0.728
Marigold-Light (Ke et al., 2025)	0.546 [.462,.645]	0.154 [.124,.187]	0.392 [.333,.459]	1.148
CRefNet (Luo et al., 2024)	0.151 [.138,.164]	0.201 [.165,.241]	**0.355 [.309,.406]**	0.484
Ordinal Shading (Careaga and Aksoy, 2023)	0.252 [.229,.279]	0.148 [.130,.166]	0.549 [.463,.637]	0.924

is significant in each case (Sect. 5.6 reports the intervals). On the hue-stability axis in isolation, two external methods are genuinely more invariant than our model.

Second, that ranking partly reflects what those methods discard. On the primary calibration guard $\text{Chroma}_{\text{err}}$, which asks whether each material matches MID's pseudo-GT hue rather than merely a stable one, our full model scores 0.121 and *every* external method is significantly worse: CRefNet 0.201, Marigold-App 0.195, Marigold-Light 0.154, Ordinal 0.148. The collapse detector explains the gap directly: CRefNet retains under half the pseudo-GT between-material chroma spread ($\text{Chroma}_{\text{fid}} = 0.484$) and Marigold-App under three quarters (0.728), whereas our model retains 94.1% of it (0.941). A prediction cannot drift in a dimension it has thrown away, so the two methods that lead Cast_{rel} lead it partly by not carrying the colour that could vary.

The defensible claim is therefore narrow and, we think, stronger for it: our model is the only one in the comparison that is best on lightness stability (C_{mat}, significant over every other row), best on colour calibration ($\text{Chroma}_{\text{err}}$, significant over every external method), and within a fifth of the most invariant method on hue. Relative to Marigold this uses roughly 70× fewer trainable parameters and about one fifth of the inference time; relative to CRefNet it uses 3.6× fewer trainable parameters and is about 3× faster (Table 5.1). It is not the best calibrated of our *own* exploratory refinement rows (two loss-refinement variants reach $\text{Chroma}_{\text{err}} = 0.115$); the refinement that buys its lightness-stability and MAW advantages costs a little chroma calibration, and we report that trade rather than hide it. The ablation of Sect. 5.6 demonstrates the mechanism inside our own architecture: a configuration that collapses chroma to $\text{Chroma}_{\text{fid}} = 0.55$ (almost exactly CRefNet's figure) is likewise flattered by the superseded pooled score of Eq. (5.2), and is visibly desaturated to the eye.

Figure 5.5 makes that last point directly, because it is the one claim in this chapter that can be checked without reference to any metric at all. Placed beside the pseudo-GT albedo, the two predictions with the lowest pooled chroma cast, our own colour-path-off configuration and CRefNet, are visibly drained of colour, while the full model more closely follows the reference. The $\text{Chroma}_{\text{err}}$ and aggregate spread ratio printed beneath each column quantify those properties. A metric that ranks the first two of those images above the third is not measuring colour constancy.

Figure 5.4 plots stability against pseudo-GT calibration for the roster. Lower is better on both axes. CRefNet and Marigold-App reduce drift more than our model, but at substantially higher calibration error; the colour-path-off ablation occupies the same high-error regime. The plot therefore exposes a trade-off rather than defining a single winner.

Figure 5.6 shows a qualitative comparison. Each column is a different illuminant condition for the same scene. Our model keeps the wall and floor colours consistent across columns; Marigold-App shows a visible hue shift between the warm-light and cool-light conditions.

5.3 IIW: Perceptual Reflectance Ordering

IIW (Bell et al., 2014) has been the default benchmark of the field for a decade, so a decomposition method is expected to report it, and we do. It scores a prediction with the Weighted Human Disagreement Rate (WHDR): human annotators are shown two points in a photograph and asked which surface is lighter, and WHDR is the confidence-weighted fraction of those comparisons that the predicted albedo orders the wrong way,

$$\text{WHDR} = \frac{\sum_i w_i \cdot \mathbb{1}\left[\hat{r}_i \neq r_i\right]}{\sum_i w_i}, \tag{5.8}$$

where r_i is the human verdict on the i-th pair (lighter, darker, or equal), $\hat{r}_i$ is the verdict implied by the predicted albedo under a threshold on the ratio of the two

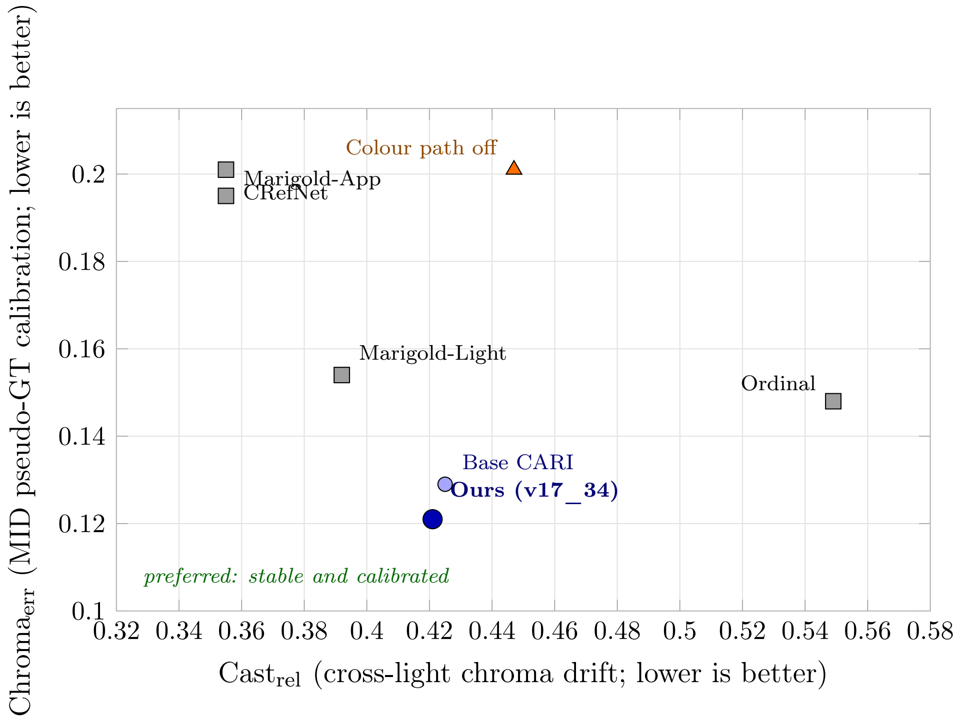

Figure 5.4: *Cross-light chroma stability against pseudo-GT chroma calibration on the 30-scene MID test split. Lower is better on both axes. Marigold-App and CRefNet have lower drift but substantially higher calibration error; the selected model occupies a compromise region rather than winning either property universally. The companion spread diagnostic in Figure 5.5 shows whether low drift is accompanied by chroma collapse.*

points' intensities, and w_i is the annotator confidence. Lower is better.

Eq. (5.8) is worth writing out because its limitations are visible in it. The comparison $\hat{r}_i$ depends only on the *ratio* of two predicted intensities, so WHDR is invariant to any global rescaling of the albedo and, more importantly, it never looks at colour at all: the albedo is reduced to a lightness ordering before it is scored. A model can therefore tint every surface in the image with the colour of the lamp and lose nothing on this metric, as long as the relative lightness of surface pairs survives. Its labels are sparse and perceptual, resting on human judgements rather than a physical albedo measurement. Subjectivity is not inherently a weakness: when the intended property is perceived relative reflectance, people are the appropriate observers. The limitation is one of *scope*. A sparse ordinal lightness label cannot determine absolute intensity, chromaticity, or texture, so complementary metrics are needed for those properties (Wu et al., 2023; Sato et al., 2025).

This matters directly for the claim of this thesis. Coloured illuminant leakage

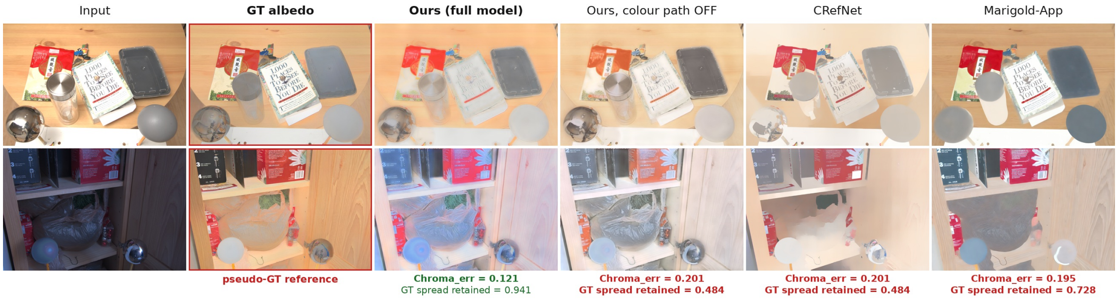

Figure 5.5: *Chroma diagnostics on two distinct MIDIntrinsics test scenes and illuminant conditions (one row each). From left: the input, pseudo-GT albedo (red border), and four predictions. Albedo is defined only up to a global scale, so every panel is independently scale-normalised for display; no colour correction is applied, so the differences in saturation are differences in the predictions themselves. Values below each column are measured over the full 30-scene test split: Chroma$_{err}$ is the pseudo-GT calibration guard and the aggregate spread ratio diagnoses collapse. The colour-path-off row and CRefNet both retain* 48.4% *of pseudo-GT chroma spread and have error 0.201, while the selected model retains* 94.1% *with error 0.121.*

Figure 5.6: *Qualitative albedo constancy on one MID scene. The upper row shows four inputs and the pseudo-ground-truth albedo; the lower row shows the corresponding predictions and a per-pixel variation heatmap (red = the prediction changes with the light). The predicted material colours stay stable across illuminants, yielding cool (blue) CoV maps on the flat surfaces where the light changes most.*

is a *hue* error, and Eq. (5.8) is hue-blind by construction, so WHDR cannot confirm or refute the property we are trying to improve. We therefore report it as complementary evidence: it tests whether enforcing colour constancy damages the relative reflectance ordering that viewers care about, but it cannot validate the coloured-illumination claim by itself. The benchmark that does address material colour directly is MAW, which Sect. 5.4 turns to next and which was created for exactly this gap (Wu et al., 2023).

Table 5.3 separates locally evaluated rows from cited full-split context. Supervision status is stated explicitly because the local block contains both zero-shot models and models trained on IIW. The cited numbers are not used as a head-to-head comparison with our local rows: each source has its own inference implementation and training protocol.

The *Ours (IIW-fine-tuned)* row makes the mismatch concrete rather than rhetorical. Fine-tuning our full model on the IIW training split with the ordinal-hinge loss does what it is asked to do: WHDR falls from 0.264 to 0.220. The accuracy and lightness-constancy guards regress in exchange, all past the pre-specified 10% no-regression threshold. The damage is heaviest on *lightness*: MID C_{mat} rises from 0.130 to 0.191 (+47%) and MAW intensity error from 0.624 to 1.057 (+69%). This is the expected signature because WHDR is an ordinal *lightness* score and optimising it reshapes lightness stability first. But it is not confined there: MAW ΔE, measured against *physically measured* albedo, worsens from 3.981 to 5.163 (+30%), so colour accuracy degrades too, even though the in-domain MID hue-drift and chroma-collapse guards remain comparatively stable ($\text{Chroma}_{\text{err}}$ $0.121 \rightarrow 0.134$, $\text{Chroma}_{\text{fid}}$ $0.941 \rightarrow 0.978$, and Cast_{rel} $0.421 \rightarrow 0.426$), which reminds us that the external measured reference is the stricter test. The shape of the trade differs from the chroma collapse by which the feed-forward baselines reach low chroma cast (Sect. 5.2). There the casualty is hue; here it is chiefly lightness. The lesson is nevertheless identical: improving one scalar benchmark can hide failures that only complementary guards expose. We report the fine-tuned row for completeness and as this demonstration, not as the selected model: the WHDR gain is real, but it does not serve our stated objective because it is bought at the expense of the constancy the thesis aims to protect.

Figure 5.7 shows qualitative results on the IIW test set, and illustrates the mismatch between what the metric rewards and what this thesis is optimising.

5.4 MAW: Measured Real-World Albedo

MAW (Wu et al., 2023) was built to close the gap that Sect. 5.3 just described. Instead of asking people which of two patches looks lighter, its authors recover a physical albedo value for real surfaces. They photograph a surface twice, once with a grey card of known reflectance resting on it and once without. Both the surface and the card receive the same illumination, so the shading term is shared and cancels in the ratio of the two measurements, which leaves the surface's albedo in absolute terms. The ground truth is therefore a measured quantity rather than a perceptual vote, which makes MAW the benchmark best able to answer the question this thesis poses: when a real room is lit by a coloured light, does the predicted material colour match the material's real colour?

Table 5.3: *Results on IIW (Bell et al., 2014). WHDR is lower-is-better. The upper block is evaluated locally; rows using IIW supervision are identified explicitly. The first cited block is selected no-IIW-supervision rows from Ordinal Shading (Careaga and Aksoy, 2023), Table 2. The second is the full-IIW roster reproduced by Wang et al. (Wang et al., 2025), Table I, an arXiv preprint that evaluates all 5,230 IIW images with per-method recommended settings. Repeated methods are intentional: they expose source- and protocol-dependent reported values, not a unified rerun. Both cited blocks are context rather than a head-to-head comparison with our local split. † denotes IIW fine-tuning: the* Ours (IIW-fine-tuned) *row is our full model fine-tuned on the IIW training split (the complement of this test split, so no leakage), and its WHDR is therefore not zero-shot. Its number is* italicised, not bolded, *because it is bought at a cost the WHDR column cannot see: fine-tuning for WHDR degrades MID lightness constancy (C_{mat}) by* 47% *(Table 5.2 baseline* 0.130 → 0.191*), which is precisely the antagonism Sect. 5.2 predicts and the reason we do not adopt it as the headline model.*

Model	WHDR ↓
Ours (full model)	0.264
Ours (base CARI)	0.286
Ours (IIW-fine-tuned)†	*0.220*
Marigold-App (Ke et al., 2025)	0.193
Marigold-Light (Ke et al., 2025)	0.215
CRefNet (local; IIW-supervised) (Luo et al., 2024)	**0.168**
Ordinal (local) (Careaga and Aksoy, 2023)	0.257
Cited from Ordinal Shading, Table 2 (no IIW supervision):	
Bell et al. (Bell et al., 2014)	0.206
CGIntrinsics (Li and Snavely, 2018a)	0.178
NIID-Net (Luo et al., 2020)	0.166
PIE-Net (Das et al., 2022)	0.213
Ordinal Shading (Careaga and Aksoy, 2023)	0.249
Full IIW test split, Wang et al., Table I:	
Bell et al. (Bell et al., 2014)	0.206
CGIntrinsics (Li and Snavely, 2018a)	0.154
NIID-Net (Luo et al., 2020)	0.166
CRefNet (Luo et al., 2024)	0.145
Texture-aware IID (Wang et al., 2025)	0.136
Original CRefNet result (different protocol):	
CRefNet† (Luo et al., 2024)	0.108

Because that ground truth is measured rather than derived from the prediction, MAW scores colour *correctness* against an external reference, not merely stability

Figure 5.7: *Qualitative albedo predictions from our model on IIW test images. Each row shows an input image (left) and its predicted albedo (right). The predictions preserve broad material regions but show the two competing errors that an ordinal score can expose only at annotated point pairs: legitimate material boundaries may be over-smoothed, while illumination structure remains on the ceiling and around the windows. This figure is an honest failure case rather than evidence of colour accuracy, because IIW provides no measured RGB albedo.*

across conditions. Both metrics remove the global-scale ambiguity: ΔE is computed on chromaticity, while the intensity error is explicitly scale-invariant. MAW therefore measures the right *chromaticity* and the right *relative* brightness, not an absolute radiometric value. It is thus complementary to MID rather than a substitute for it: a model can be perfectly stable across illuminants (a low C_{mat}) and still be consistently wrong about what colour the wall is, and MAW is what catches that.

The public scorer first resizes the prediction and region mask to 320×240 and extracts a robust median RGB vector $\hat{\mathbf{a}}_m$ for each measured material region m; $\mathbf{a}^\star_m$ is the corresponding measured reference and w_m is the region area. Let $\ell(\mathbf{x}) = \frac{1}{3}\sum_c x_c$ denote the scorer's RGB channel mean. The colour score removes intensity independently for each material, $\beta_m = \ell(\mathbf{a}^\star_m)/\ell(\hat{\mathbf{a}}_m)$, and reports the area-weighted CIEDE2000 colour difference,

$$\Delta E_{00} = \frac{\sum_m w_m \, \Delta E_{00}(\mathrm{Lab}(\beta_m \hat{\mathbf{a}}_m), \mathrm{Lab}(\mathbf{a}^\star_m))}{\sum_m w_m}. \tag{5.9}$$

This per-material scale alignment makes the score primarily a chromaticity test rather than a test of recovered intensity. It is therefore the primary external score for a coloured-illuminant leak.

The second score tests relative material brightness. Writing $\hat{l}_m = \ell(\hat{\mathbf{a}}_m)$ and $l^\star_m = \ell(\mathbf{a}^\star_m)$, the scorer fits one area-weighted global scale

$$\alpha^\star = \frac{\sum_m w_m \hat{l}_m l^\star_m}{\sum_m w_m \hat{l}_m^2}, \qquad \text{SI-MSE} = \frac{\sum_m w_m (\alpha^\star \hat{l}_m - l^\star_m)^2}{\sum_m w_m}. \tag{5.10}$$

Because the scale is fitted away, this score is insensitive to a uniformly too-dark or too-bright prediction and instead penalises the wrong *relative* lightness of different materials. It does not directly measure spatial blur, although smoothing can change the robust region medians on which it is computed.

We run every local method with the common 1280-pixel long-side inference cap specified in Sect. 5.1; the public scorer then performs its fixed 320×240 evaluation and scale handling, matching the MAW protocol used by CD-IID (Careaga and Aksoy, 2024).

Table 5.4 reports ΔE, the CIELAB colour difference between the prediction and measured albedo, and intensity SI-MSE$\times$100, which removes the global-scale ambiguity before assessing brightness. Lower is better for both. The cited block is retained only for methods reported under the same MAW protocol, with CD-IID Table 1 as the primary external reference.

The IIW-fine-tuned diagnostic confirms the trade-off with an independent measured reference. Its 17% WHDR improvement is accompanied by a 30% increase in MAW ΔE and a 69% increase in intensity error. The effect is therefore not confined to MID's dataset-derived pseudo-ground truth: direct optimisation of the sparse ordinal objective also damages colour and relative intensity against physically measured albedo.

Figure 5.8 shows a representative MAW scene. The ground-truth panel is masked to the surface regions whose albedo was physically measured by the dataset authors; only those regions enter the score.

5.5 ARAP: Constancy and Albedo Accuracy

ARAP (Bonneel et al., 2017) is synthetic, and we report it last for that reason: a rendered scene is an easier target than a photograph, so a good ARAP number cannot by itself support a claim about real rooms. What ARAP uniquely offers is that it is the only benchmark here with dense ground-truth albedo *and* several renderings of the same scene under different lights. MID has the repeated illuminants but no true albedo; MAW has the true albedo but only one illuminant per surface. ARAP has both, so it is the one place where constancy and accuracy can be measured on the very same pixels, and where a claim that improving one damages the other can actually be tested.

It answers two distinct questions, which must not be merged. First, does a prediction stay stable when the original coloured illumination changes? This is tested on the raw renderings with our self-consistency measures (Table 5.5). Second, how accurate is the recovered albedo when the input is neutrally lit? This is tested on white-balanced input against dense ground-truth albedo (Table 5.6).

Table 5.4: *Results on MAW (Wu et al., 2023), evaluated with the benchmark scorer. ΔE measures CIELAB colour difference and intensity SI-MSE$\times$100 measures scale-invariant brightness error; lower is better for both. The upper block is evaluated locally. The complete lower block is transcribed from CD-IID (Careaga and Aksoy, 2024), Table 1. † marks the diagnostic full model after IIW fine-tuning; it is included to expose cross-benchmark damage and is not the selected model. * denotes the grayscale-shading assumption, for which predicted chromaticity is inherited from the input image. Gold, silver, and bronze cells mark first, second, and third place over the common protocol.*

Model	$\Delta E \downarrow$	Int. SI-MSE$\times$100 $\downarrow$
Ours (full model)	3.981	0.624
Ours (IIW-fine-tuned)†	5.163	1.057
Ours (base CARI)	4.155	0.504
Ours (colour path only; Table-A Row 3)	4.115	**0.395**
Marigold-App (Ke et al., 2025)	3.775	0.653
Marigold-Light (Ke et al., 2025)	4.214	0.493
CRefNet (Luo et al., 2024)	3.970	0.942
Ordinal (local) (Careaga and Aksoy, 2023)	6.884	0.657
Cited from CD-IID, Table 1:		
Bell et al. (Bell et al., 2014)	6.61	3.11
CGIntrinsics (Li and Snavely, 2018a)	5.15	2.71
Li & Snavely (2018a)* (Li and Snavely, 2018b)	6.56	1.72
NIID-Net (Luo et al., 2020)	4.73	1.24
IRISformer (Zhu et al., 2022b)	4.94	1.44
Kocsis et al. (Kocsis et al., 2024)	5.35	1.13
IntrinsicAnything (Chen et al., 2024)	4.12	0.98
Ordinal Shading* (Careaga and Aksoy, 2023)	6.56	0.57
CD-IID (Careaga and Aksoy, 2024)	**3.37**	0.54

We white-balance correctly, which is worth stating because an earlier version of our evaluator did not. A single global exposure rescale (multiplying all three channels by one scalar) leaves the chromaticity of the illuminant untouched and is therefore *not* white balance; measured on a coloured-lit ARAP scene it leaves the implied illuminant as far from neutral after the operation as before. We instead reconstruct an achromatic-illuminant image from the ground-truth albedo: writing the implied shading as $\mathbf{S} = \mathbf{I}/\mathbf{A}^*$, we recompose $\mathbf{I}_{\text{wb}} = \mathbf{A}^* \cdot \text{lum}(\mathbf{S})$, which drives the implied illuminant chromaticity to exactly neutral. The accuracy table uses this reconstruction; the constancy table uses the raw coloured renderings, since removing the illuminant would erase the very variable the constancy measure exists to probe.

Figure 5.8: *Qualitative comparison on three content-diverse MAW photographs (one per row): a kitchen, a dark office, and a living area. Left to right: input photograph, measured albedo (red border and masked to reliable regions), and the prediction from Ours. All three middle panels show the corresponding physically measured MAW albedo mask.*

Two coverage points follow. The constancy table (Table 5.5) is restricted to the multi-illuminant groups whose illuminant *colour* actually changes between renderings (16 groups overall, 11 indoor): cross-light drift is undefined for a single-light scene, and groups that vary only intensity or direction carry no colour cast for an invariant albedo to be tested against, so including them would pull every model's score toward zero and mask the effect the thesis is about. The accuracy table (Table 5.6) instead scores every image with usable dense albedo, namely 136 of the 157 in the collection; the other 21 have near-black ground truth (fewer than 88 valid pixels) and are dropped deterministically for every model alike. The accuracy table combines our local white-balanced evaluation with published original-ARAP results from Ordinal Shading (Careaga and Aksoy, 2023), Table 1.

Figure 5.9 provides the model-by-illuminant comparison that the aggregate tables cannot show spatially. Figure 5.10 then samples three distinct ARAP interiors to reduce dependence on that single scene.

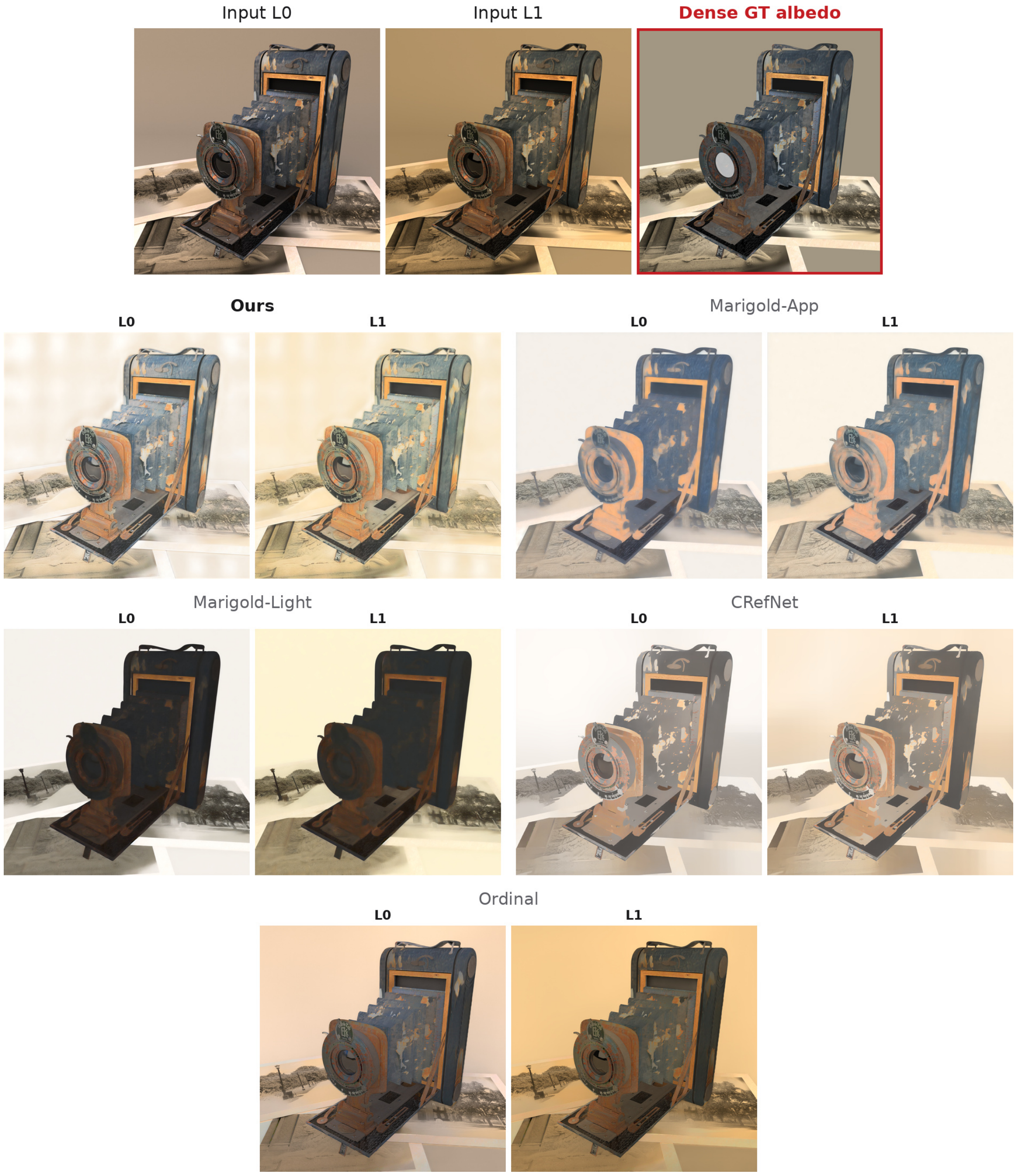

Figure 5.9: *Model-by-illuminant comparison on the ARAP* camera *scene. The reference strip shows both rendered illuminant conditions and the dense ground-truth albedo once. The grid below shows each locally evaluated model's albedo under the same two illuminants, so the reader can compare consistency without repeatedly duplicating the reference. Predictions are independently scale-normalised for display, with no colour correction. The row labelled Ours uses the model selected for the thesis qualitative figures. The grid exposes different failure modes rather than declaring a winner: CRefNet suppresses much of the material colour, Marigold-Light and Ordinal retain illumination structure, and our L1 output still contains hard-shadow leakage. Marigold-App is comparatively stable on this scene; the full-set metrics in Table 5.5 and Table 5.6 remain the aggregate evidence.*

Table 5.5: *ARAP raw-input constancy on the* colour-varying *groups of the original ARAP collection (Bonneel et al., 2017) — the groups in which the illuminant* colour *changes between renderings, which is the literal case this thesis is about (*16 *groups overall,* 11 *indoor; the remaining multi-light groups vary only intensity/direction and are excluded, since they carry no colour for the albedo to be invariant to and would dilute the score toward zero).* C_{arap} *measures cross-illumination albedo variation and* $Cast_{RMS}$ *residual chroma-cast variation; lower is better for every column. Gold, silver, and bronze mark first, second, and third numerical rank in each column; displayed ties share a rank. These are internal diagnostics, evaluated locally.*

	All colour-varying (n=16)		Indoor colour-varying (n=11)	
Model	C_{arap} ↓	Cast_{RMS} ↓	C_{arap} ↓	Cast_{RMS} ↓
Ours (full model)	0.142	*0.079*	**0.096**	*0.077*
Ours (base CARI)	0.154	*0.092*	0.119	*0.094*
Marigold-App (Ke et al., 2025)	**0.113**	***0.052***	0.113	***0.038***
Marigold-Light (Ke et al., 2025)	0.340	*0.197*	0.340	*0.256*
CRefNet (Luo et al., 2024)	0.128	*0.122*	0.129	*0.137*
Ordinal Shading (Careaga and Aksoy, 2023)	0.175	*0.294*	0.165	*0.355*

Cast_{RMS} is shown in *italics*. Its rank colours indicate numerical order only, not a stand-alone quality claim. Unlike the MID pooled score it carries no between-material term. It is a chroma variance taken across the illuminants of a single scene, but it remains an *absolute* variance, which a colourless albedo minimises just as effectively as an invariant one. It is reported for completeness and read against the chroma-fidelity figures of Table 5.2: the methods that lead this column are the same ones that retain the least albedo chroma. C_{arap} is also a pure constancy score and can be minimised by a constant prediction; it must be read beside the dense-GT accuracy guard in Table 5.6.

Table 5.6: *ARAP albedo accuracy,* within-study, *on the full Bonneel et al. collection (Bonneel et al., 2017) (136 of 157 images have dense ground-truth albedo with enough valid pixels to score; the same 136 are used for every model). Input is white-balanced by reconstructing an achromatic-illuminant image from the ground-truth albedo (Sect. 5.5). LMSE and mean-normalised RMSE are lower-is-better, SSIM higher. The upper block is evaluated locally under one harness; the lower block is transcribed from Ordinal Shading (Careaga and Aksoy, 2023), Table 1. Gold, silver, and bronze mark first, second, and third place across all rows.*

Model	LMSE ↓	mn-RMSE ↓	SSIM ↑
Ours (full model)	0.0275	<u>0.2150</u>	<u>0.8083</u>
Ours (base CARI)	0.0279	0.2259	0.8067
Marigold-App (Ke et al., 2025)	0.0266	0.2529	0.8016
Marigold-Light (Ke et al., 2025)	0.0706	0.6187	**0.8204**
CRefNet (Luo et al., 2024)	0.0223	0.2916	0.8051
Published original-ARAP protocol (Ordinal Shading, Table 1):			
Chromaticity (Bonneel et al., 2017)	0.024	0.296	0.692
NIID-Net (Luo et al., 2020)	0.022	**0.206**	0.781
Li & Snavely (Li and Snavely, 2018b)	**0.018**	0.227	0.749
Bell et al. (Bell et al., 2014)	0.028	0.317	0.718
PIE-Net (Das et al., 2022)	0.026	0.273	0.754
Ordinal Shading (Careaga and Aksoy, 2023)	<u>0.021</u>	0.252	0.761

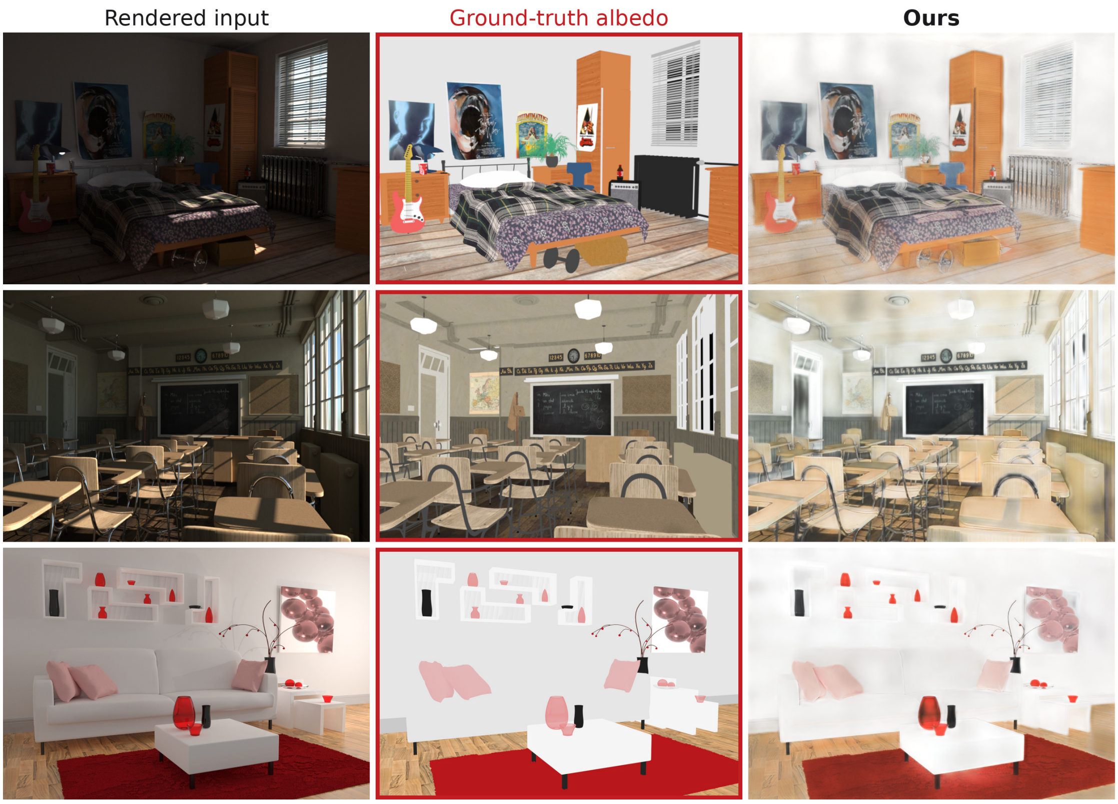

Figure 5.10: *Qualitative accuracy on three content-diverse ARAP scenes:* bedroom, classroom, *and* livingroom. *Each row shows the rendered input, dense ground-truth albedo (red border), and the prediction from Ours. This figure tests scene diversity and dense-albedo fidelity; Figure 5.9 separately compares models while the illuminant changes.*

5.6 Ablation Study

The benchmark sections placed our model against external methods. We now turn inward with one matched CARI study and one exploratory refinement study. The first, the *CARI ablation* of this section, asks whether the constancy our model achieves genuinely comes from CARI, or from something incidental; it isolates the two CARI components in a 2×2 ablation. The second, the *refinement study* of Sect. 5.7, takes the resulting full-CARI model and asks whether it can be improved further, by targeted extra loss terms or by the cross-illuminant training data of Sect. 4.2.4. Together they explain both *why* the model is built the way it is and *what* the cheaper alternatives cost.

The ablation isolates the contribution of CARI by comparing configurations along two axes: whether the cross-render losses ($\mathcal{L}_{\mathrm{inv}}$, $\mathcal{L}_{\mathrm{expl}}$) are active, and whether the colour-path bundle (wide RGB skip and its chroma-loss setting) is active (Sect. 3.2). These are the two components that, according to the coupling argument in that section, are expected to interact. Table 5.7 summarises the resulting 2×2 design. Part 1 reports self-consistency on raw-coloured inputs: the MID measures with their fidelity guard, then ARAP constancy over all eligible groups and over the domain-matched indoor subset. Part 2a reports measured-albedo accuracy, the GT-achromatized ARAP accuracy guard, and ancillary IIW ordering. Part 2b reports raw-input ARAP accuracy over the same all-ARAP and indoor subsets as Part 1. The GT-achromatized ARAP columns support the within-study comparison but are not presented as a reproduced published score. All four configurations share the same 19k initialization, data curriculum, and 40k training horizon. The two within-architecture comparisons are the causal CARI tests: Row 1→Row 2 toggles cross-render losses with the colour path off, and Row 3→Row 4 does the same with the colour path on. These two within-architecture contrasts are the matched CARI tests. The cross-axis colour-path comparison is not causal: it changes head parameters and chroma supervision, starts Adam afresh, and the recorded fallback produces an approximately 1.85× different effective learning-rate trajectory. We therefore report cross-axis differences descriptively rather than attributing them to the skip alone.

This ablation is also where the defect in the pooled chroma score of Eq. (5.2) becomes visible, and we present it in that light because the correction set out in Sect. 5.2 was in fact discovered here. Under Eq. (5.2), Row 2 is the best configuration in the table because it attains the lowest pooled cast of the four, while Row 4, the full model, is the worst. Inspection of the predicted albedo contradicts this immediately: Row 2's albedo is visibly desaturated, and Row 4's is not. The decomposition of Eq. (5.3) resolves the contradiction. Row 2 retains only 0.484 of the pseudo-GT between-material chroma spread, so most of its low pooled score is the *absence of colour*, not the absence of drift. Once the drift is measured against the chroma each model actually carries (Eq. (5.5)), the ordering reverses and Row 4 becomes the most invariant configuration of the four.

Effect of cross-render losses within each architecture. The clean CARI comparisons are Row 1→Row 2 and Row 3→Row 4, the two contrasts in which only the cross-render losses change. The clearest effect is on *lightness* stability: cross-render supervision lowers C_{mat} from 0.250 to 0.180 with the colour path off and from 0.259 to 0.157 with it on, corresponding to a 28% and 39% reduction, both well outside the scene-bootstrap intervals of Table 5.2. On scale-normalised hue invariance the effect is present but small: $\mathrm{Cast}_{\mathrm{rel}}$ moves from 0.477 to 0.447 and from 0.444 to 0.425. The relevant paired bootstrap, rather than overlap of marginal intervals, gives Row 1-minus-Row 2 0.030 with a 95% interval $[0.011, 0.049]$, and Row 3-minus-Row 4 0.019 with interval $[0.007, 0.031]$. Both are modest but significant hue-stability gains.

The pseudo-GT calibration guard supplies the necessary qualification. With the colour path off, $\mathrm{Chroma}_{\mathrm{err}}$ worsens from 0.186 to 0.201; with it on, the change from 0.124 to 0.129 is not significant. CARI therefore improves paired constancy but does not itself improve colour calibration. It also improves ancillary IIW WHDR in the first pair, while both MAW measures worsen slightly. The load-bearing result is the repeated paired constancy gain, read beside rather than instead of the external accuracy guards.

Association with the colour-path bundle. The colour-path rows retain 92.5% and 99.9% of the pseudo-GT aggregate chroma spread, while the rows without it retain 51.2% and 48.4%. They also have lower $\mathrm{Chroma}_{\mathrm{err}}$ and stronger MAW colour, with Row 3 obtaining the best ΔE (4.115) and intensity score (0.395). This is strong descriptive evidence that the bundle preserves colour. It is not a skip-only causal estimate because the bundle changes chroma supervision, head parameters, optimizer state, and effective learning-rate history.

This is the opposite of what the pooled score reports, and the discrepancy is instructive rather than awkward. Under Eq. (5.2) the colour-skip rows appear to *double* their chroma cast (0.287 to 0.519, 0.268 to 0.555); under Eq. (5.3) that increase is almost entirely the between-material term $S(A)$ rising from roughly 0.25 to roughly 0.48 as the model recovers the chromatic diversity that the ground-truth albedo actually possesses ($S(A^*) = 0.495$). The pooled metric was charging the model for the colour it restored relative to the pseudo-reference. Row 4 is consequently the strongest configuration in the table on both constancy axes, C_{mat} and $\mathrm{Cast}_{\mathrm{rel}}$, and is not the weakest, contrary to what the superseded score indicated.

Table 5.7: *CARI study across cross-render losses and the colour-path bundle. All configurations share the same 19k initialization and data curriculum, and 40k horizon; Row 1→Row 2 and Row 3→Row 4 are the matched cross-render comparisons. Cross-axis comparisons are descriptive because the colour path also changes chroma supervision, optimizer state, and effective learning-rate history. Part 1 reports raw-input constancy on the colour-varying ARAP groups (the same* 16/11 *subset used in Table 5.5 and Table 5.8); Part 2a measured-albedo accuracy, the GT-achromatized ARAP guard and ancillary IIW ordering; Part 2b raw-input ARAP accuracy. Lower is better everywhere except ARAP SSIM, and the aggregate spread ratio, whose target is* 1.000. *Best per column is* ***bold****. Figure 5.11 shows the corresponding albedo and variation maps.*

Part 1: Raw-input constancy (self-consistency across illuminants)

			MID (30 scenes)				ARAP all (col. varying, n=16)		ARAP indoor (col. varying, n=11)	
Row	Cross-render	Colour path	$C_{\mathrm{mat}} \downarrow$	$\mathrm{Cast}_{\mathrm{rel}} \downarrow$	$\mathrm{Chroma}_{\mathrm{err}} \downarrow$	Spread ratio ($\to 1$)	$C_{\mathrm{arap}} \downarrow$	$\mathrm{Cast}^{\S}_{\mathrm{RMS}}$	$C_{\mathrm{arap}} \downarrow$	$\mathrm{Cast}^{\S}_{\mathrm{RMS}}$
1	off	off	0.250	0.477	0.186	0.512	0.164	0.102	0.122	0.105
2	on	off	0.180	0.447	0.201	0.484	**0.145**	*0.080*	**0.108**	*0.085*
3	off	on	0.259	0.444	**0.124**	0.925	0.155	0.106	0.117	0.105
4	on	on	**0.157**	**0.425**	0.129	**0.999**	0.154	0.092	0.119	0.094

MID spread ratios use the ratio of aggregate predicted and pseudo-GT spreads, not the biased mean of per-scene ratios. The ARAP $\mathrm{Cast}_{\mathrm{RMS}}$ columns carry no between-material term but are still *absolute* chroma variances, which a colourless albedo also minimises.

Part 2a: Standard-protocol accuracy and structure (against ground truth)

			MAW		ARAP (GT-achromatized)			IIW
Row	Cross-render	Colour path	$\Delta E \downarrow$	Int.×100 ↓	LMSE† ↓	mn-RMSE† ↓	SSIM† ↑	WHDR ↓
1	off	off	4.434	0.409	0.0307	0.2378	0.8055	0.334
2	on	off	4.631	0.474	0.0320	0.2313	**0.8069**	**0.283**
3	off	on	**4.115**	**0.395**	0.0283	0.2348	0.8047	0.327
4	on	on	4.155	0.504	**0.0279**	**0.2259**	0.8067	0.286

Part 2b: Raw-input ARAP albedo accuracy (against ground truth)

			All ARAP (raw input)		Indoor ARAP (raw input)	
Row	Cross-render	Colour path	LMSE ‡ ↓	mn-RMSE ‡ ↓	LMSE ‡ ↓	mn-RMSE ‡ ↓
1	off	off	0.0463	0.341	0.0441	0.323
2	on	off	0.0397	**0.292**	**0.0370**	**0.275**
3	off	on	0.0426	0.319	0.0400	0.301
4	on	on	**0.0396**	0.293	0.0382	0.276

† GT-achromatized ARAP accuracy guard (136 of 157 images): an internal within-study comparison, not a reproduced literature table. IIW WHDR is ancillary and is not a selection criterion. ‡ Raw-input ARAP accuracy is a local multi-illuminant stress test and is not compared with published white-balanced results.

Figure 5.11: *Ablation: the four CARI configurations on one MID scene under three illuminant conditions (L0, L10, L20). Columns: the three input conditions, the pseudo-ground-truth albedo (red box), the three predicted albedos, and two cross-condition variation heatmaps computed across five illuminant conditions (L0, L5, ..., L20; 5, three of which are shown): luminance CoV, matching MID C_{mat}, and the legacy pooled chroma-cast CoV diagnostic $Cast_{RMS}$ ($\sqrt{\mathrm{Var}(R/G) + \mathrm{Var}(B/G)}$ across conditions). Rows, top to bottom: no CARI (Row 1), cross-render only (Row 2), colour-skip only (Row 3), and full CARI (Row 4). These are local, single-scene illustrations rather than estimates of the full-split metrics in Table 5.7: the heatmaps expose where the predicted albedo changes with illumination, while the table establishes the dataset-level ranking. In particular, chroma variation is spatially and scene dependent, so the figure is not used to infer the global $Cast_{RMS}$ ordering.*

Role of IIW and ARAP in the next study. IIW WHDR is a sparse perceptual ordinal measure; it is therefore retained as a complementary column rather than the colour-constancy selection criterion. Its Row 2 result is consistent with the real cross-render signal helping ordinal reflectance, but does not establish that transfer on its own. ARAP varies less across the four rows than MID: all-ARAP C_{arap} spans 0.145–0.164 and indoor-ARAP 0.108–0.122, while the corresponding MID C_{mat} range is 0.157–0.259. This limited ARAP response motivates testing a synthetic cross-render source in the next ablation; it is a hypothesis motivated by the measured domain difference, not a conclusion about its cause.

Why Row 4 initializes Table B. Row 4 is used as the Table-B starting point because it is the complete CARI configuration and, on the corrected metrics, the strongest of the four rows on both constancy axes while retaining 99.9% of the pseudo-GT aggregate chroma spread. Its residual weaknesses (a slightly higher $\mathrm{Chroma}_{\mathrm{err}}$ than Row 3 (0.129 against 0.124) and a marginally worse MAW ΔE (4.155 against 4.115)) remain explicit in Table 5.7 and inform the next study's ARAP–MAW selection criteria.

Figure 5.11 shows the qualitative effect of each ablation row.

5.7 Refinement Study: Loss Levers versus Cross-Illuminant Data

The ablation established that CARI works. This study asks the natural follow-up: can the full-CARI model be pushed further, and if so, by *what kind* of intervention? Two candidates are available, and they are of fundamentally different types. The first is a pair of extra *loss terms* ("levers"), each targeting a residual weakness visible in the full-CARI albedo: smooth shading ramps that survive on flat surfaces, and shadow boundaries that leak into the albedo. The second is more *data*, specifically the 3D-Front-IID corpus (Sect. 4.2.4), which supplies the wide, deliberately coloured illuminant changes that neither Hypersim nor MID provides. Running both in one controlled study lets us ask a question neither could answer alone: if the levers are compensating for a signal missing from the training data, then supplying that signal should reduce or remove the need for them.

The two loss levers. Writing $\mathbf{A}$ and $\mathbf{S}_d$ for the predicted albedo and diffuse shading and $\mathbf{I}$ for the input:

- **Flatness** (*flat*): a chroma-gated low-frequency albedo-flatness prior. With $\mathbf{A}_{\mathrm{LF}}$ a low-pass of the predicted albedo and w_c a per-pixel gate that is high only where the local chromaticity is uniform (same material), $\mathcal{L}_{\text{flat}} = \sum_p w_c(p)\,\|\nabla \mathbf{A}_{\mathrm{LF}}(p)\|_1$. It removes smooth shading ramps left in the albedo while the chroma gate preserves genuine material and texture edges.

- **Shadow-invariance** (*sh_inv*): a manufactured-shadow self-supervision. A smooth synthetic relighting field σ is cast onto a Hypersim image, and the predicted albedo is required to be invariant to it: $\mathcal{L}_{\text{sh\_inv}} = \|\mathbf{A}(\mathbf{I}\odot\sigma) - \mathbf{A}(\mathbf{I})\|_1$.

The synthetic relighting used by the shadow lever is shown in Figure 5.12. This branch is separate from CARI: CARI's $\mathcal{L}_{\text{inv}}$ uses real MID raw pairs, whereas the shadow lever manufactures a second observation on Hypersim with a *known* field and applies only $\mathcal{L}_{\text{sh\_inv}}$. The distinction matters for the reading below: the shadow lever is a synthetic stand-in for exactly the kind of second observation that 3D-Front-IID provides for real.

Design. Every row of Table 5.8 starts from the same full-CARI checkpoint (Row 4 of Table 5.7 at 40k iterations) and trains for a further 20k iterations to a 60k horizon, at the same constant learning rate. The rows share initialization, horizon, and learning rate, but the logged data mixtures are not identical. Rows 1–2 use Hypersim/MID/InteriorVerse weights 0.4/0.3/0.3; Row 3 changes mixture during

Figure 5.12: *Synthetic relighting used by the shadow-invariance lever. A known multiplicative field is applied to a Hypersim image to manufacture a shadowed or sunlit second observation, and the predicted albedo is required to stay invariant under it. This loss is distinct from CARI's real-pair invariance loss, and it is the manufactured counterpart of the genuine cross-illuminant pairs supplied by 3D-Front-IID (Sect. 4.2.4).*

training before ending at 0.3/0.3/0.15/0.25 with 3D-Front-IID; Row 4 uses that standardized mix; and Row 5 uses approximately 0.32/0.24/0.24/0.20 throughout. Current YAML files do not retroactively define these checkpoints, so this provenance comes from the logs. Row 1 is the zero-lever control: it receives the identical 20k iterations of extra training with neither intervention, so every other row is read against it rather than against the base checkpoint, and the effect of the extra training itself is not mistaken for the effect of a lever or a dataset. Row 2 adds the two levers. Row 3 instead adds 3D-Front-IID, with both lever weights explicitly zero, isolating the data axis. Rows 4 and 5 add levers, but their recorded mixtures differ. Consequently only Rows 1→2 is a matched lever contrast. The 3D-Front and flatness comparisons are exploratory associations rather than single-variable ablations. A no-regression gate, specified in the training configuration before the runs, applies throughout: an intervention is kept only if it does not regress the coloured-illuminant constancy metrics or the albedo-accuracy guard by more than approximately 10% relative to the control. We describe it as pre-specified rather than pre-registered: it is fixed in the config, not lodged as a timestamped artefact ahead of the results.

What the comparison can establish. Rows 1→2 establish the effect of the two levers under the no-3D-Front mixture. The remaining rows support model selection and hypothesis generation, not component attribution, because their mixture histories differ. The selected configuration is therefore the best observed gate-passing trade-off, reported alongside the full-CARI starting point rather than replacing it silently.

Illuminant invariance does not move. The first result of the study is a negative one, and it is reported first because it constrains how everything else may

Table 5.8: *Refinement study. All rows resume the same full-CARI checkpoint (Table 5.7, Row 4, at 40k) and train a further 20k iterations to 60k at the same learning rate. Only Rows 1→2 is a matched contrast; the 3D-Front rows have the logged mixture histories stated above and are interpreted exploratorily. Lower is better on every column. Guards are not used for selection: the gate is applied to the constancy columns and MAW* ΔE.

			MID			ARAP indoor (col. varying)		MAW		Guards	
Row	Configuration	Levers	$C_{\text{mat}}\downarrow$	$\text{Cast}_{\text{rel}}\downarrow$	$\text{Chroma}_{\text{err}}\downarrow$	$C_{\text{arap}}\downarrow$	$\text{Cast}_{\text{RMS}}\downarrow$	$\Delta E\downarrow$	Int.×100 ↓	mn-RMSE ↓	WHDR ↓
1	Control (extra training only)	–	0.161	0.415	**0.115**	0.116	0.090	4.179	**0.511**	0.2194	0.283
2	+ levers	flat, sh_inv	0.146	0.426	**0.115**	0.100	0.085	4.193	0.732	**0.2099**	0.281
3	+ 3D-Front-IID	–	0.163	0.418	0.120	0.111	0.080	4.040	0.512	0.2274	0.280
4	+ 3D-Front + levers	flat (reduced), sh_inv	**0.130**	0.421	0.121	**0.096**	**0.077**	**3.981**	0.624	0.2150	**0.264**
5	+ 3D-Front + levers	flat (full), sh_inv	0.149	0.421	0.143	0.097	0.078	4.231	0.719	0.2103	0.282

Guard columns (ARAP mn-RMSE, IIW WHDR) are reported for transparency and are not selection criteria: IIW WHDR is a sparse perceptual ordinal measure with a different target from colour constancy (Sect. 5.3). Cast_{rel} is deliberately *not* bolded: its five values span 0.415–0.426, a range of 2.6%, and no row is meaningfully more illuminant-invariant than any other. That column is a null result and is discussed as such.

be read. Neither intervention changes illuminant invariance. Cast_{rel} takes the values 0.415, 0.426, 0.418, 0.421 and 0.421 across the five rows, a spread of 2.6%, with the zero-lever control among the best and the two combined rows indistinguishable from it. Whatever the levers and the 3D-Front-IID corpus contribute, they do not make the albedo more stable under a change of illuminant than twenty thousand further iterations of ordinary training would have.

This is worth stating plainly because the superseded pooled score would have suggested otherwise. Under Eq. (5.2) the same five rows span 0.523 to 0.586, a 12% range, and appear to rank cleanly. That apparent ranking is the between-material term of Eq. (5.3), not drift: Row 5 (the full-strength flatness prior) posts the worst pooled cast of the five (0.586) purely because it is the most over-saturated row in the table; its aggregate spread ratio is 1.055, while Row 4 retains 0.941. The pooled score was ranking the rows partly by how much albedo chroma each carried. Once that term is removed, the constancy axis is flat, and the study must be adjudicated on the axes that do respond.

What does respond: chroma fidelity, lightness stability, and measured colour. The rows are not inert; they move calibration, lightness stability, and measured colour. Row 4 has the best lightness stability ($C_{\text{mat}} = 0.130$), MAW colour ($\Delta E = 3.981$), coloured-illuminant ARAP scores (0.096/0.077), and IIW ordering (0.264). Its MID $\text{Chroma}_{\text{err}} = 0.121$ is slightly worse than the 0.115 controls, and its MAW intensity score is also worse, so selection is a trade-off rather than a clean sweep.

Row 5 is associated with worse pseudo-GT calibration ($\text{Chroma}_{\text{err}} = 0.143$), lightness stability (0.149), and MAW ΔE (4.231) than Row 4. Because its data mix also differs, this cannot isolate the full flatness weight as the cause. The matched Rows 1→2 comparison shows that the levers improve C_{mat} and the ARAP guard while worsening MAW intensity; Row 3 shows that the 3D-Front configuration can improve MAW colour without improving MID lightness stability. The combined Row 4 is retained as the best observed compromise, not as proof of complementary causal effects.

Row 4 is therefore the selected configuration and is the model reported as *Ours* throughout Sect. 5.2–Sect. 5.5.

6 | In Practice

The previous chapter measured the decomposition; this chapter puts it to use. Separating a photograph into an illuminant-invariant albedo and a coloured shading layer is not only an estimation target but a practical handle: once the two factors are available, an image can be altered by changing one of them and leaving the other untouched, which no edit on the entangled pixels can do. This chapter develops that idea into a single application, *factor-space editing*, and presents two instances of it that follow directly from the properties established in Chapter 5.

This chapter is a proof of concept, not a second evaluation. Each edit is defined directly on the predicted factors, and its behaviour follows from how those factors are constructed, so the argument is made by formulation rather than by a quantitative application study, for which no user study or application-specific metric is available. Every factor comes from the single trained model selected in Sect. 5.5, labelled simply *Ours*; the chapter draws no comparison between internal checkpoints. That model's decomposition is shown first (Figure 6.1), and the two edits are then read as operations on its predicted albedo, shading, and residual.

All edits below start from the decomposition $\mathbf{I} = \mathbf{A}_d \odot \mathbf{S}_d + \mathbf{R}$, where $\mathbf{A}_d$ is the predicted diffuse albedo, $\mathbf{S}_d$ the coloured diffuse shading, and $\mathbf{R} = (\mathbf{I} - \mathbf{A}_d \odot \mathbf{S}_d)_+$ the analytic non-diffuse residual that captures specular highlights, visible light sources, and clipped regions (Sect. 3.1).

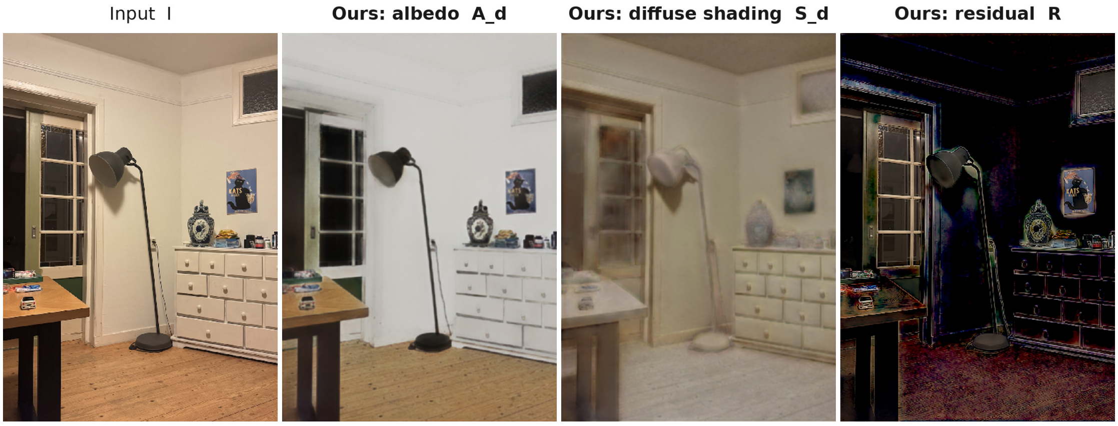

Figure 6.1: *Predicted decomposition by Ours on a realistic indoor photograph. From left: input photograph; predicted diffuse albedo $\mathbf{A}_d$; diffuse shading $\mathbf{S}_d = (1-\boldsymbol{\pi})/\boldsymbol{\pi}$ obtained by inverting the model's inverse-domain shading output $\boldsymbol{\pi}$; and the analytic residual $\mathbf{R} = (\mathbf{I} - \mathbf{A}_d \odot \mathbf{S}_d)_+$. Albedo and shading are independently normalised for display. The residual is shown at its calculated magnitude, without amplification, so it remains dark except at bright non-diffuse regions such as glass and highlights. This is a qualitative factorisation, not ground truth. The four panels are direct crops from the saved visualization; no new inference or post-hoc colour correction is applied for this figure.*

We do not include a cross-illuminant transfer demonstration, in which one scene is re-rendered under another scene's illuminant. Its natural score, the PSNR of the recomposed image, is highest for the visibly desaturated colour-path-off ablation, because compensating albedo and shading errors cancel in their product; it is a gameable factor-consistency diagnostic rather than evidence of physical correctness. The two edits below instead each act on a single factor of the decomposition, one on the residual $\mathbf{R}$ and one on the shading $\mathbf{S}_d$, so their behaviour follows directly from how those factors are defined.

6.1 Factor-Space Editing

Residual-space suppression. The residual layer contains image energy not represented by the predicted diffuse product. This can include specular highlights and clipped sources, but it can also include ordinary scene structure that the estimated albedo and shading fail to reconstruct. Scaling it yields the edit

$$\mathbf{I}'_\alpha = \mathbf{A}_d \odot \mathbf{S}_d + \alpha\,\mathbf{R}, \qquad \alpha \in [0,1]. \tag{6.1}$$

Setting $\alpha < 1$ dims energy assigned to the residual while holding the model's predicted diffuse product fixed; $\alpha = 0$ removes that component entirely. The same operation performed in image space (for example, lowering the exposure of bright regions) also changes diffuse regions; residual scaling instead holds the model's predicted diffuse product fixed. This is the same operation used for highlight editing by Careaga and Aksoy (2024), but whether it behaves as a selective highlight control depends on the quality of the decomposition.

Because $\mathbf{R}$ is the analytic leftover of Figure 6.1 rather than a segmented highlight region, reducing α suppresses everything the diffuse product fails to explain, diffuse surfaces included, not one isolated highlight. Residual scaling is therefore a targeted edit on a predicted factor, not a semantic glare mask or a general-purpose glare remover; how selective it appears is governed entirely by the quality of the decomposition.

Illumination editing. Classic brightness or contrast adjustments operate on the entangled image, so lifting dark regions also washes out material texture and colour. With the decomposition, tonal edits can be applied to the shading layer alone:

$$\mathbf{I}'_f \;=\; \mathbf{A}_d \odot f(\mathbf{S}_d) + \mathbf{R}, \tag{6.2}$$

where f is a monotone luminance gain. Because our shading is unbounded, we do not apply $\mathbf{S}_d^{\gamma}$ directly. We normalise shading luminance by its 95th percentile q_{95} and use

$$f_\gamma(\mathbf{S}_d) = \mathbf{S}_d \left[\mathrm{clip}\left(\frac{L(\mathbf{S}_d)}{q_{95}}, \varepsilon, 1\right)\right]^{\gamma-1}, \qquad 0 < \gamma < 1, \tag{6.3}$$

with the gain capped at 2.5. Low predicted illumination is lifted while values at or above the reference percentile are unchanged. Because $\mathbf{A}_d$ is untouched, the model's predicted material colours and texture are held fixed while its predicted illumination changes. This is where the invariance targeted throughout the thesis pays off directly: an albedo that does not carry the illuminant's colour can be re-lit without its materials drifting in hue, so lifting a dark, warm-lit corner reveals the surfaces' own colour rather than amplifying the lamp's cast. The result is a shading edit built directly on the decomposition, and its fidelity is bounded by how completely that cast was removed in the first place.

By construction this holds the predicted albedo of Figure 6.1 fixed, whereas the same luminance gain applied to the entangled image multiplies material and illumination together and so shifts material appearance as well. The edit is therefore a statement about controllability of the predicted factors, not about ground-truth preservation.

Both edits succeed exactly to the degree the decomposition does, and their two failure modes map onto the two axes measured in Chapter 5. Wherever some illuminant colour has leaked into the albedo (the residual cast quantified by the MID and ARAP constancy metrics), that component is already baked into the material layer and will not respond to a shading edit. And wherever the shading is structurally soft (a sharp shadow boundary that was never fully separated from the albedo), scaling the shading cannot cleanly remove it. Neither limitation is repaired by the editing operation; each is inherited from the factors being edited, which is why the constancy and structure results of the previous chapter are what ultimately bound how far this application can be taken.

7 Conclusion

7.1 Summary

This report presented an intrinsic image decomposition model trained with Cross-Render Albedo Invariance (CARI). The central problem we addressed is the leakage of coloured illumination into the predicted albedo: when a room is lit by a warm yellow lamp, standard IID models tend to predict a slightly yellow version of the wall's true colour because they have no mechanism to distinguish the lamp's hue from the material's hue.

CARI attacks this problem at the training stage rather than the model stage. By presenting the model with pairs of images of the same scene photographed under different coloured lights, and penalising any change in the predicted albedo between the two images, we force the model to assign all the illuminant colour information to the shading layer. This approach requires two practical ingredients: a dataset with real multi-illuminant pairs (we use MID), and a commitment to not applying white balance to those pairs. White balance neutralises the illuminant colour before training, which removes the very signal that CARI needs to operate.

Establishing this required repairing the instrument before we could trust the reading. The chroma-cast score we had inherited and used in earlier drafts of this work pools albedo chromaticity across all the materials of a scene before taking its variance, and so measures the scene's ordinary chromatic diversity alongside the illuminant-induced drift it is intended to isolate. That diversity term is computed from the model's own prediction, which makes the score reducible by a model that simply desaturates its albedo. The metric, in other words, paid models to destroy the quantity intrinsic decomposition exists to recover. It ranked the configuration of our own model whose albedo is visibly drained of colour as the best in the ablation, and the colour-preserving configuration as the worst. Decomposing the score into a scale-normalised invariance term and a calibration term anchored on MID pseudo-ground truth reverses that verdict and brings the numbers back into agreement with what the images plainly show (Sect. 5.2, Figure 5.5).

Our experiments then compare CARI against a matched no-CARI baseline and against external methods across four benchmarks: MID and ARAP for colour-illuminant constancy, MAW for measured albedo accuracy, and IIW for perceptual shading quality (Chapter 5). The result is a conjunction rather than a clean sweep, and we state it as such. Our model is the only one in the comparison that is simultaneously colour-faithful and competitively invariant: it closely matches the MID pseudo-GT chroma calibration and holds the best material-lightness stability in the roster. It does *not* hold the best hue-invariance score; two external methods do. The fidelity guard shows why that ranking is not the compliment it appears to be: both of them buy their invariance by discarding real albedo colour, one of them to very nearly the same degree as the ablated configuration of our own model that we discarded on sight. Where we trail on IIW, the evidence only establishes weaker sparse relative-reflectance ordering. More decisively, fine-tuning directly for IIW lowers WHDR by 17% while worsening MID lightness constancy by 47%, MAW colour error by 30%, and MAW intensity error by 69%. The benchmark gain therefore hides damage to the constancy and measured-albedo properties central to this thesis.

The ablation identifies the reliable causal result: real cross-illuminant pairs improve constancy when the colour path is held fixed. The later refinement study is deliberately weaker evidence because its rows do not all share an identical data mixture; it records useful trade-offs without claiming that every change is caused by one isolated factor.

7.2 Limitations

Our approach has several limitations that future work could address. Figure 7.1 shows two measured failure modes directly, because most other qualitative figure in this thesis shows the method working and that is not, on its own, evidence of anything.

Constancy-structure trade-off. As described in Sect. 5.3, the model does not simultaneously achieve its best constancy and its best WHDR. The fine-tuning diagnostic shows that the two objectives can pull in opposite directions: improving sparse relative-reflectance ordering damages both material-lightness stability and measured albedo accuracy. Resolving this tension likely requires an explicit model of the illuminant, so the network can separate material colour from illuminant colour more cleanly rather than just being constrained not to let them mix.

Invariance cannot be optimised for on its own. The metric correction of Sect. 5.2 is not merely a bookkeeping fix; it constrains what any future work on

Figure 7.1: *Where the method fails. From left: the input; the ground-truth albedo; our selected model; the same model with the flatness prior at full strength; and a per-pixel map of how much our predicted albedo moves when only the illuminant moves.* ***The flatness prior is not free.*** *At full strength (Table-B Row 5) it does not merely smooth texture. It is associated with an aggregate chroma-spread ratio of* 1.055*, compared with* 0.941 *for the selected model, and costs a quarter of a* ΔE *unit on MAW. Because the recorded data mixtures differ, this does not isolate the flatness weight causally; it documents the failure of the full Row 5 configuration.* ***Non-diffuse surfaces remain unsolved.*** *The instability map is not uniform: it concentrates on the mirrored sphere and the glossy lid. An albedo–shading product has no representation for a mirror, so the residual absorbs what it can and the rest surfaces as cross-illuminant drift.*

this problem may claim. A perfectly constant predictor (one that emits flat grey albedo regardless of its input) is *perfectly* illuminant-invariant, and will score better on any pure drift measure than any model that actually recovers colour. Invariance is therefore not a quantity that can be reported, or optimised, in isolation: it is only meaningful paired with a fidelity term that is anchored on an external albedo reference rather than on the model's own output. MID provides a dataset-derived pseudo-reference; MAW supplies the independently measured check. We suspect this degeneracy is not peculiar to our formulation, since the pooled score we inherited exhibits it, and we note that two of the four external methods we evaluate lead the uncorrected score while measurably discarding albedo colour. A systematic audit of how colour constancy is scored elsewhere in the IID literature is beyond the scope of this thesis, but our result suggests it would be worth conducting.

Achromatic ambiguity. Even with perfect coloured-illuminant constancy, a different fundamental ambiguity remains: a dark textured surface (say, a dark grey carpet with visible pile) can look similar to a lighter surface under a slight shadow. This achromatic texture-shading confusion is not resolved by CARI, which only deals with coloured differences between images. It is the fundamental ill-posedness of IID, and addressing it requires geometric priors or cross-image consistency beyond what CARI provides.

Scale of multi-illuminant training data. MID contains approximately 1,000 scenes, but it is still much smaller and less varied than the synthetic corpora used for supervised IID training. The constancy constraint is therefore applied to a relatively limited set of scene-illuminant combinations, which limits how much the model can generalise. More multi-illuminant data, especially with a wider variety of illuminant colours and surface materials, would likely improve the constancy further.

Grey shading assumption in the loss, not the representation. The model's shading output is three-channel (Sect. 3.1), so it can represent spatially varying illuminant colour in principle. What remains grey-biased in practice is the *supervision*: the ground-truth shading available from Hypersim is achromatic, so the three-channel shading head is disambiguated toward colour almost entirely by CARI on MID pairs, which is a comparatively small and less spatially diverse signal. A scene with two differently coloured light sources active in different regions of the same image is not well represented in either training source, which limits how finely the model can localise a spatially mixed illuminant.

7.3 Future Work

Several directions follow naturally from this work.

Stronger and more plentiful invariance signals. The ablation points toward the signal itself: more cross-illuminant pairs, covering more materials and a wider range of illuminant colours, would directly strengthen the constraint that does the work in our approach. A structured illuminant representation (for example, a small palette of illuminant colours with a spatial assignment map) remains an interesting direction for representing *spatially mixed* illuminants, but our current evidence suggests it should be pursued as a complement to a strong invariance signal, not as a substitute for one.

Quantifying the enhancement application. Chapter 6 demonstrates that the recovered decomposition supports qualitative edits (residual suppression and factor-space illumination adjustment) without any additional training. A natural extension is to evaluate these edits quantitatively, for example by user studies on perceived realism of edited images, or downstream enhancement metrics, thereby turning the qualitative demonstration into a measured application.

Evaluation on additional coloured-illuminant benchmarks. The Beigpour MIID dataset (Beigpour et al., 2014) (Sect. 2.3) provides real scenes with multiple coloured light sources and ground-truth illuminant annotations. If access becomes available, evaluating on it would test the model on a more controlled version of the multi-illuminant problem and provide a comparison point with multi-illuminant colour-constancy methods, in addition to the IID methods compared in Chapter 5.

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